2. Solar Energy
B. Photovoltaic systems

PHOTOVOLTAIC PRINCIPLES

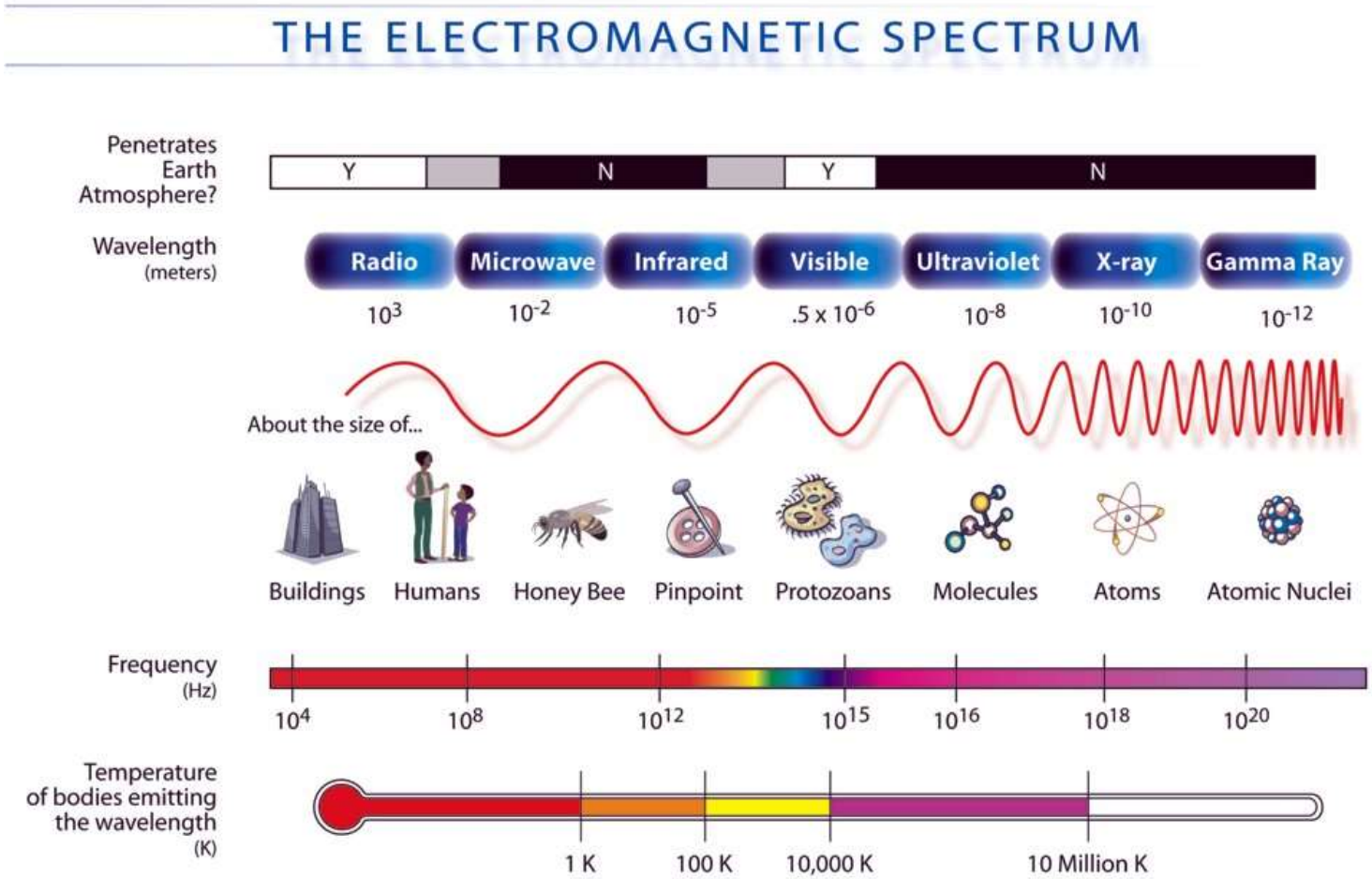
- **Photoelectric effect:** Electrons are emitted from matter under the exposure to radiation but:
  - No electron is ejected if the frequency of the light is not higher than a threshold value regardless of its intensity.
  - The electrons are ejected at even at low intensity if the frequency is high enough
  - Minimum energy required is depend on the material

  ➔➔➔ Discovery of photons (package of energy)
- Energy of photon:

  $E=h\nu$ *(h: Planck's constant (6.626×10^{-34} J·s), ν: frequency of photon)*

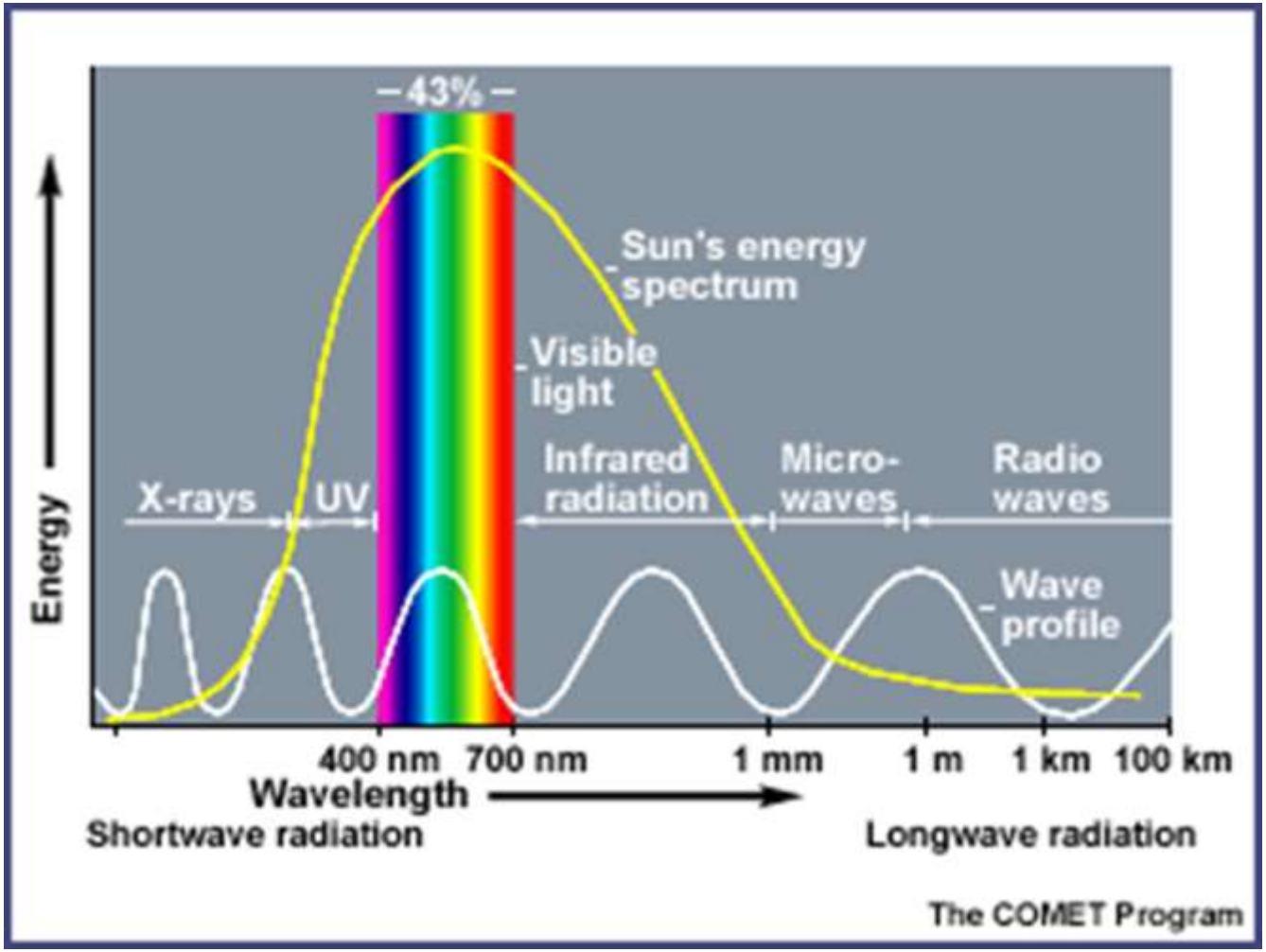
  $v=c/\lambda$ *(c: speed of light; wavelength)*

PHOTOVOLTAIC PRINCIPLES



http://mynasadata.larc.nasa.gov/ElectroMag.html

PHOTOVOLTAIC PRINCIPLES



PHOTOVOLTAIC PRINCIPLES

Photovoltaic Effect:

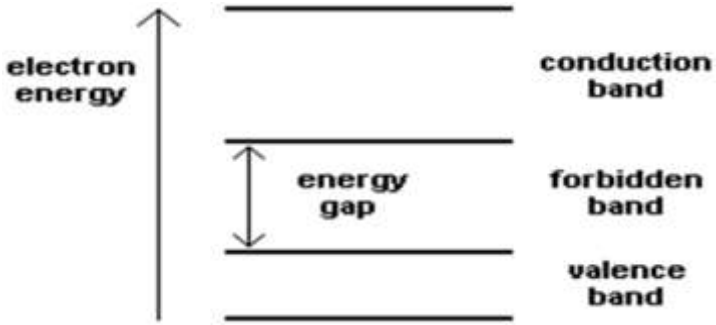
- In 1839, Edmund Becquerel, discovered that certain materials would produce small amounts of electric current when exposed to light.

WHY?

- Sunlight is composed of photons.
- Photons strike a solar material; they may be reflected or absorbed, or pass right through.
- The absorbed energy is transferred to certain electrons
- The electrons escape from its normal position to become part of the current in an electrical circuit.

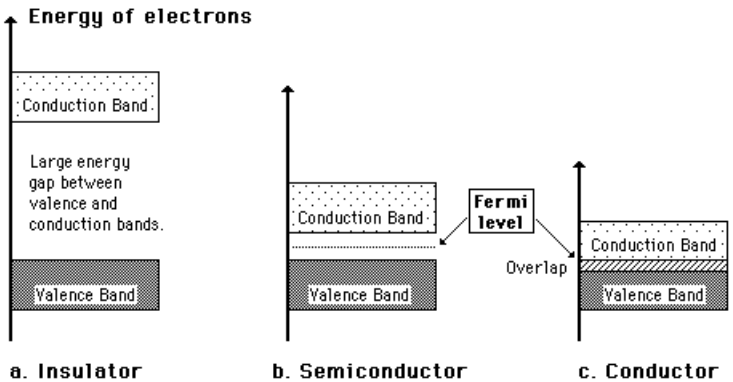
PHOTOVOLTAIC PRINCIPLES

- **Covalent bonds**: atoms in crystalline solids are held together in a regular lattice by covalent bond
- **Valence electrons:** Pair of electrons forming covalent bond.
- **Valence band:** Energy levels of valance electrons
- **Conduction electrons:** Electrons moving freely through the lattice without being held in covalent bonds
- **Conduction band:** Energy levels of conduction electrons (higher energy level than the valence band)



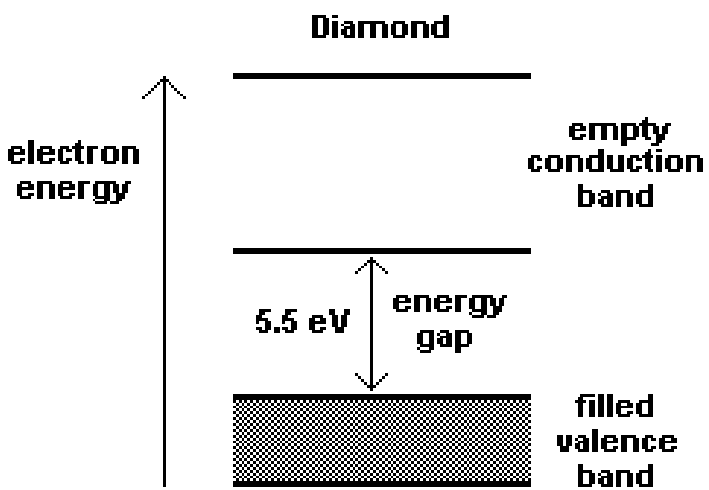
PHOTOVOLTAIC PRINCIPLES

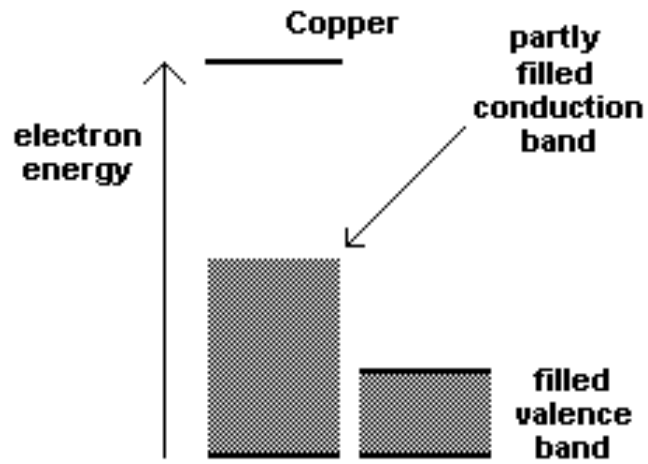
- Materials can be categorized into *conductors*, *semiconductors* or *insulators* by their ability to conduct electricity or by their "band gap".
- Semiconductor material are used to convert solar energy with electron-hole pairs.



http://people.seas.harvard.edu/~jones/es154/lectures/lecture_2/energy_gap/energy_gap.html

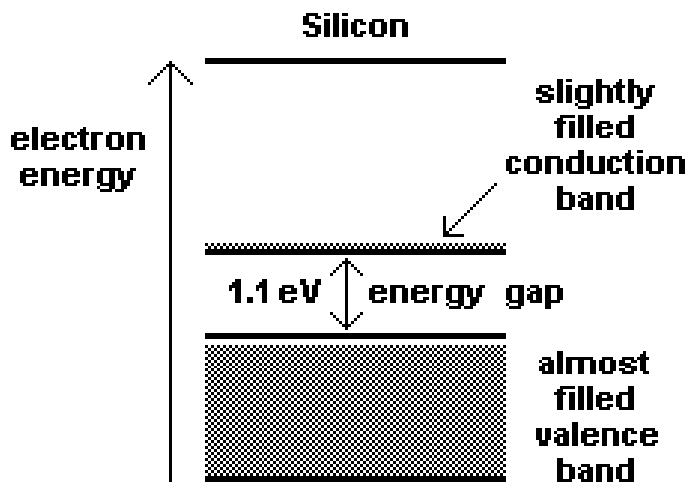
PHOTOVOLTAIC PRINCIPLES





insulator

conductor



semiconductor

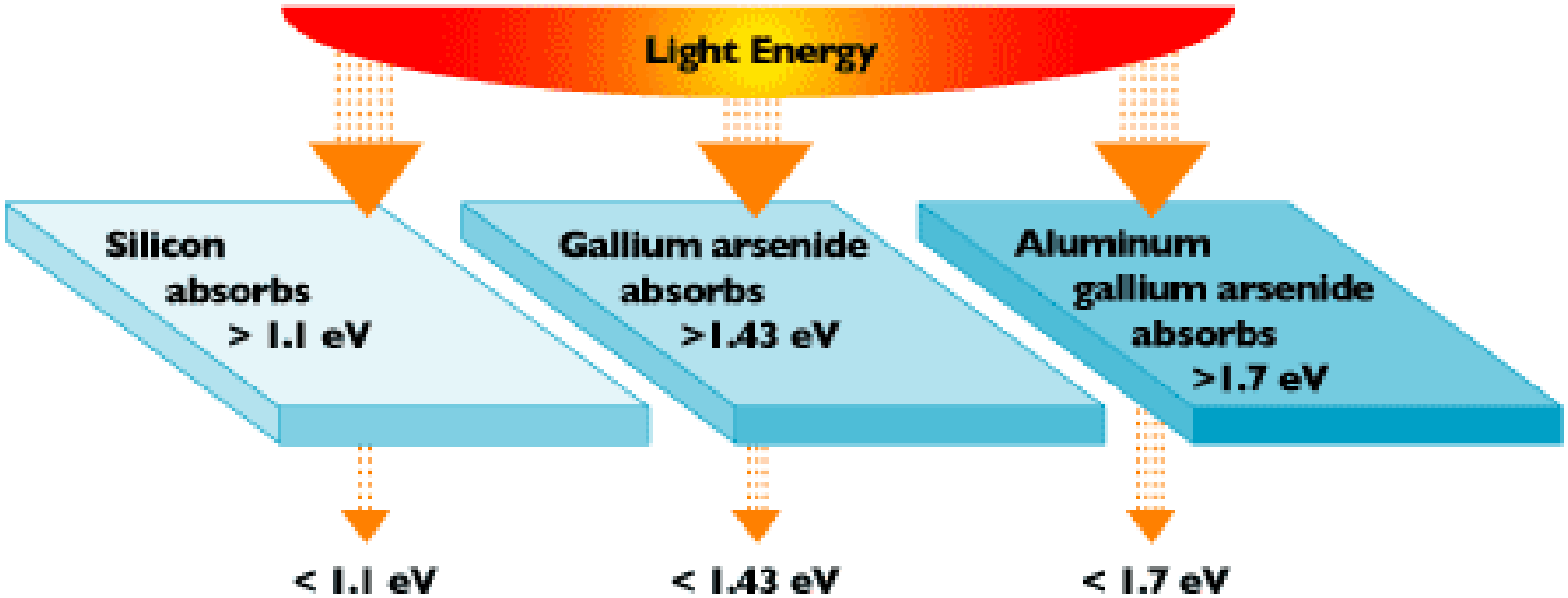
http://www.jgsee.kmutt.ac.th/exell/Solar/PVCells.html

PHOTOVOLTAIC PRINCIPLES

- When light shines on a semiconductor material, the electrons within the crystal lattice may be freed .
- The "band gap energy," is energy required to dislodge an electron from its covalent (specific to material) band
- To free an electron, the energy of a photon must be greater or equal to the band gap energy.
- Efficiency problem
  - The photons with lower energy than band gap cannot be used for electricity generation.
  - For high energy photons, only the portion of energy equal to band gap can be used

PHOTOVOLTAIC PRINCIPLES

- Crystalline silicon has a band gap energy of 1.1 electron-volts (energy gained by an electron when it passes through a potential of 1 volt in a vacuum.)
- Sunlight covers range from infrared to ultraviolet (0.5 eV to about 2.9 eV)



http://www1.eere.energy.gov/solar/bandgap_energies.html

PHOTOVOLTAIC PRINCIPLES

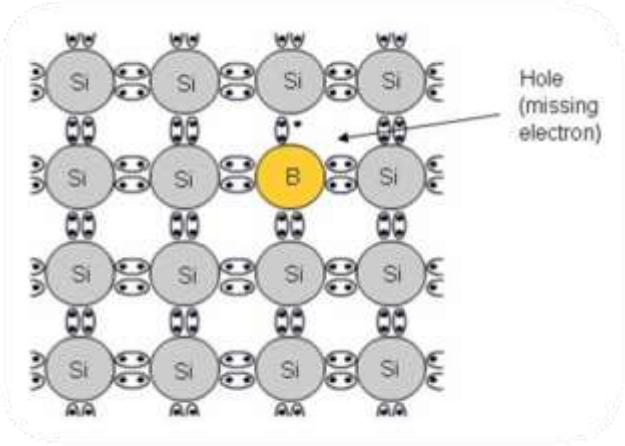
HOW IT WORKS?

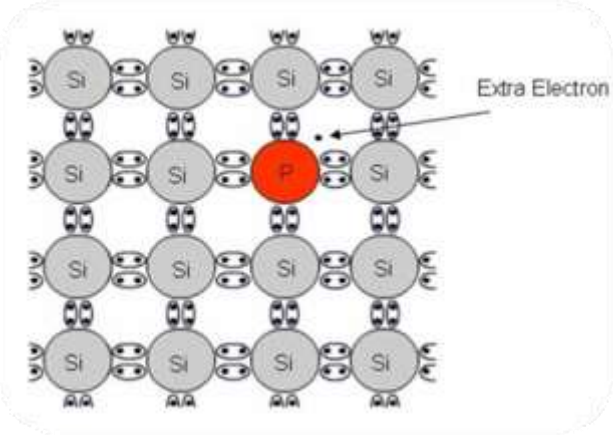
- Light shining on may free electrons within the crystal lattice. But these electrons should be directed into an electrical circuit.
- The solar cell must have **a built-in electric field.**
- In silicon based solar cells, two semiconductors are sandwiched together to build this electric field

PHOTOVOLTAIC PRINCIPLES

Built-In Electric Field

- p-type (or "positive") semiconductors have an abundance of positively charged holes,
- n-type (or "negative") semiconductors have an abundance of negatively charged electrons.
- p and n type semiconductors are sandwiched together



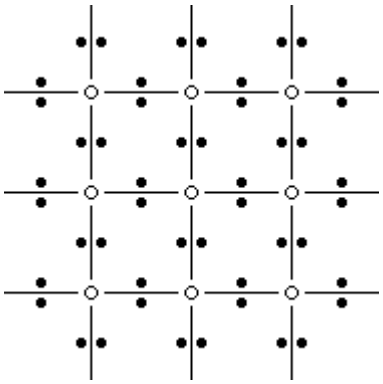


http://solarhorizon.com.au/SolarCells.html

CREATING n-TYPE and p-TYPE SILICON: "DOPING"

- n-type and p-type silicon is produced using the process called "doping", (introducing atoms of another element into the silicon crystal to alter its electrical properties).

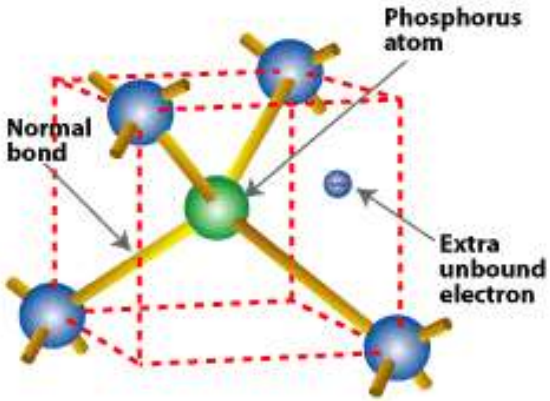
- The "dopant," (introduced element), has either three or five valence electrons (one less or more that silicon's four).



http://www.jgsee.kmutt.ac.th/exell/Solar/PVSystems.html

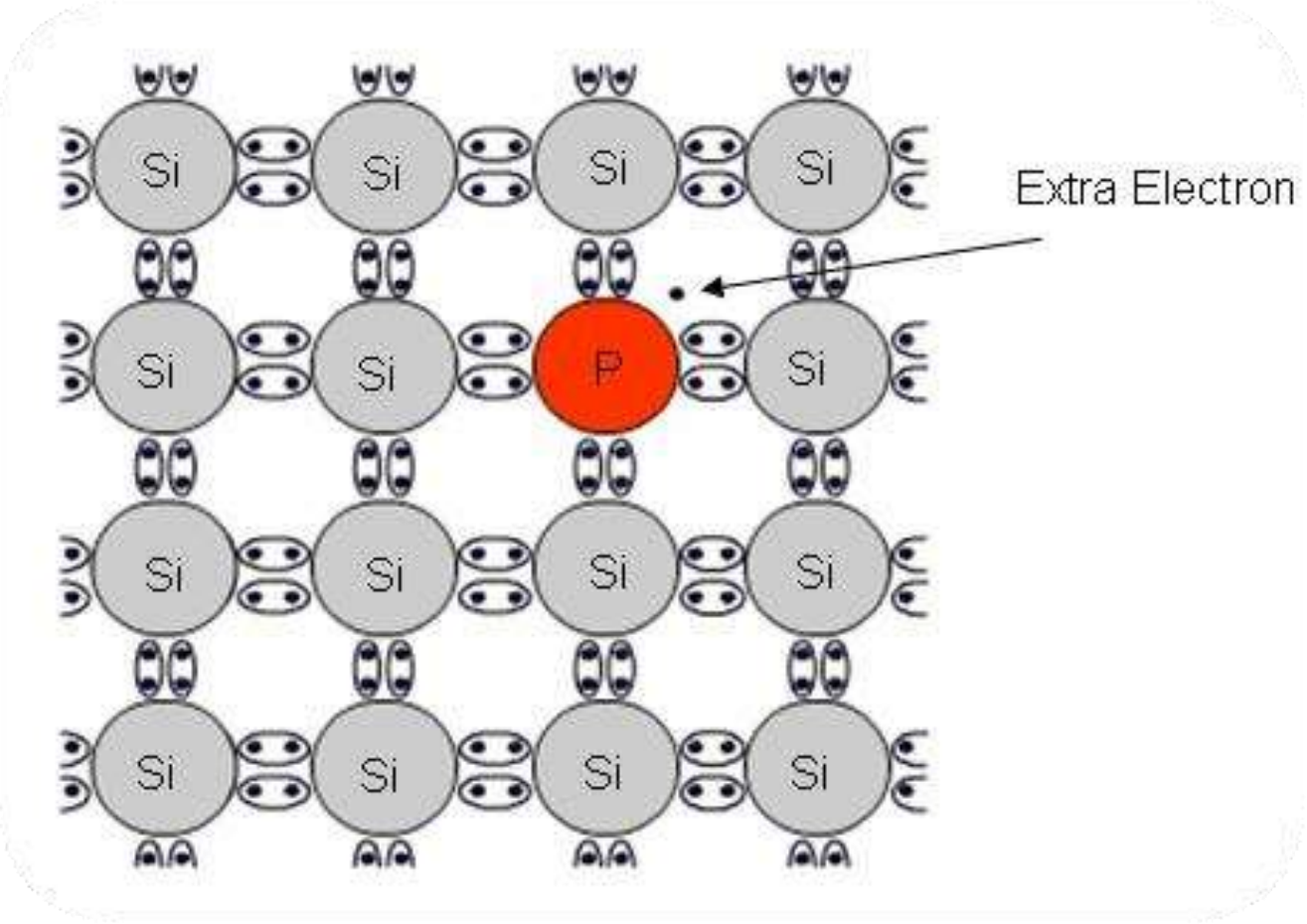
CREATING n-TYPE SILICON: P-DOPING

- Phosphorus (P) has five valence electrons, it is used for doping n-type silicon.
- P atom replaces one Si in the crystal
- Four valence electrons bond with Si atoms fifth valence electron remains free.
- The most common method of doping is to coat a layer of Si with P and heat the surface. This allows the P atoms to diffuse into the silicon.



http://www1.eere.energy.gov/solar/bandgap_energies.html

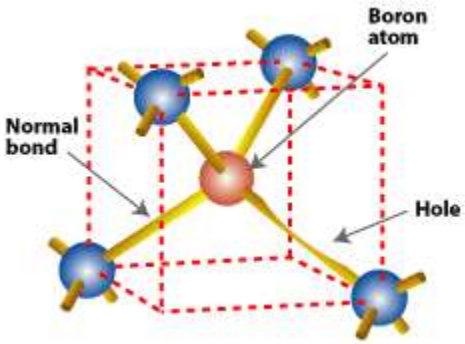
CREATING n-TYPE SILICON: P-DOPING



http://solarhorizon.com.au/SolarCells.html

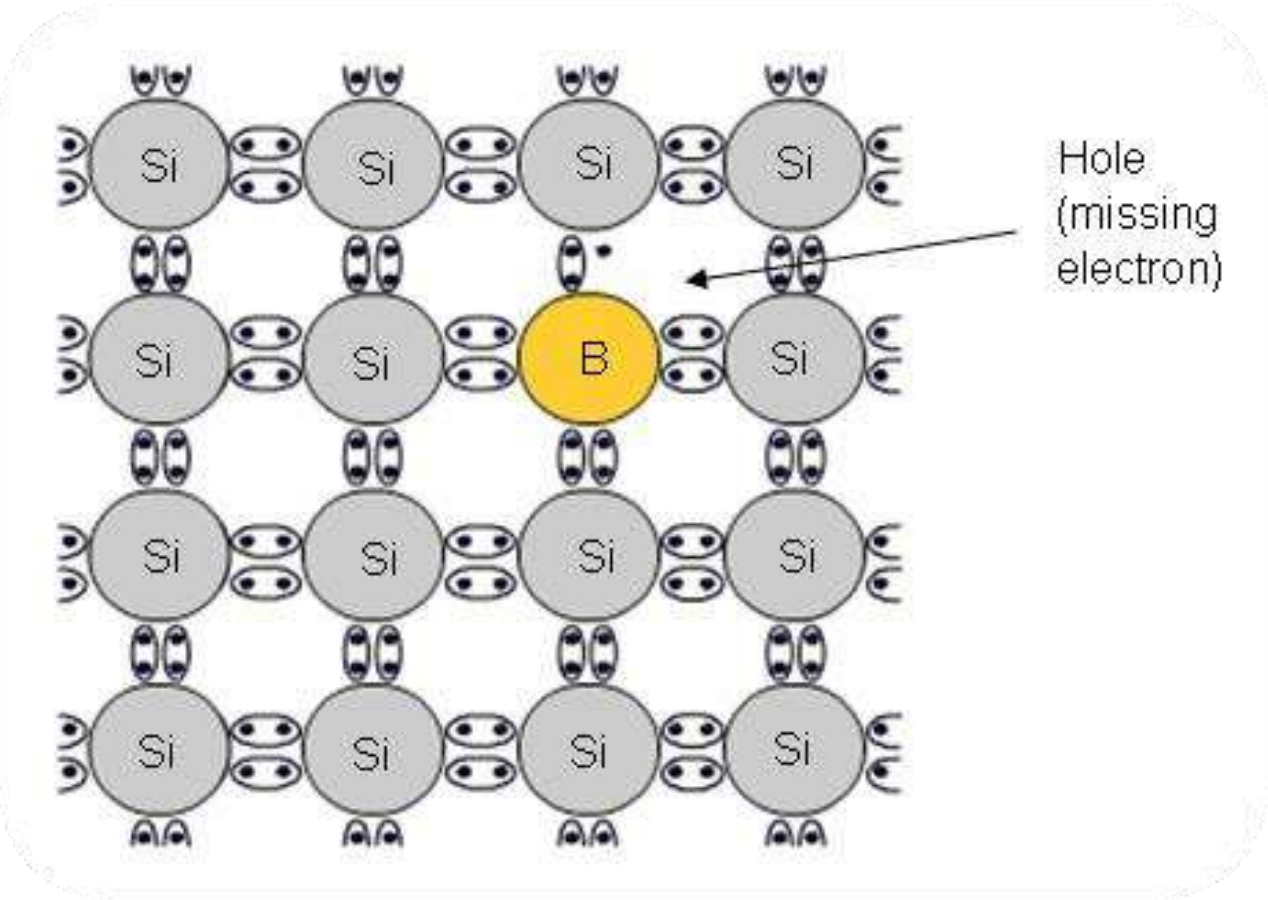
CREATING p-TYPE SILICON: B-DOPING

- Boron (B) has three valence electrons, it is used for doping p-type silicon.
- When a boron atom takes the place of a Si in crystal lattice, a bond will be missing an electron (there is an extra positively charged hole).
- Boron is introduced during silicon processing.



http://www1.eere.energy.gov/solar/bandgap_energies.html

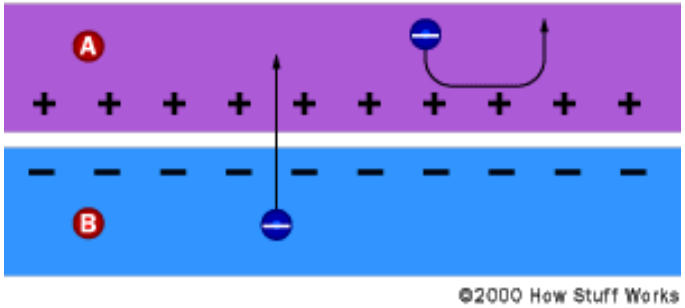
CREATING p-TYPE SILICON: B-DOPING



http://solarhorizon.com.au/SolarCells.html

PHOTOVOLTAIC PRINCIPLES

- Excess electrons in n side moves to p sites.
- A thin layer between two sides is depleted of charge carriers (electron and holes); its called depletion zone
- When equilibrium is reached, we have electric field separating two sides.
- This electric field acts as a diode, allowing (even pushing) electrons to flow from p side to n side only
- We connect two site to a circuit.
- When the photon hits the semiconductor, it moves electrons and create current



A: n-type; B: p-type
http://www.howstuffworks.com/solar-cell.htm/printable

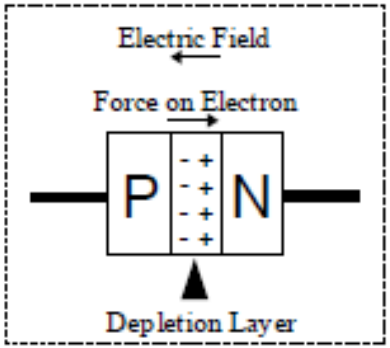
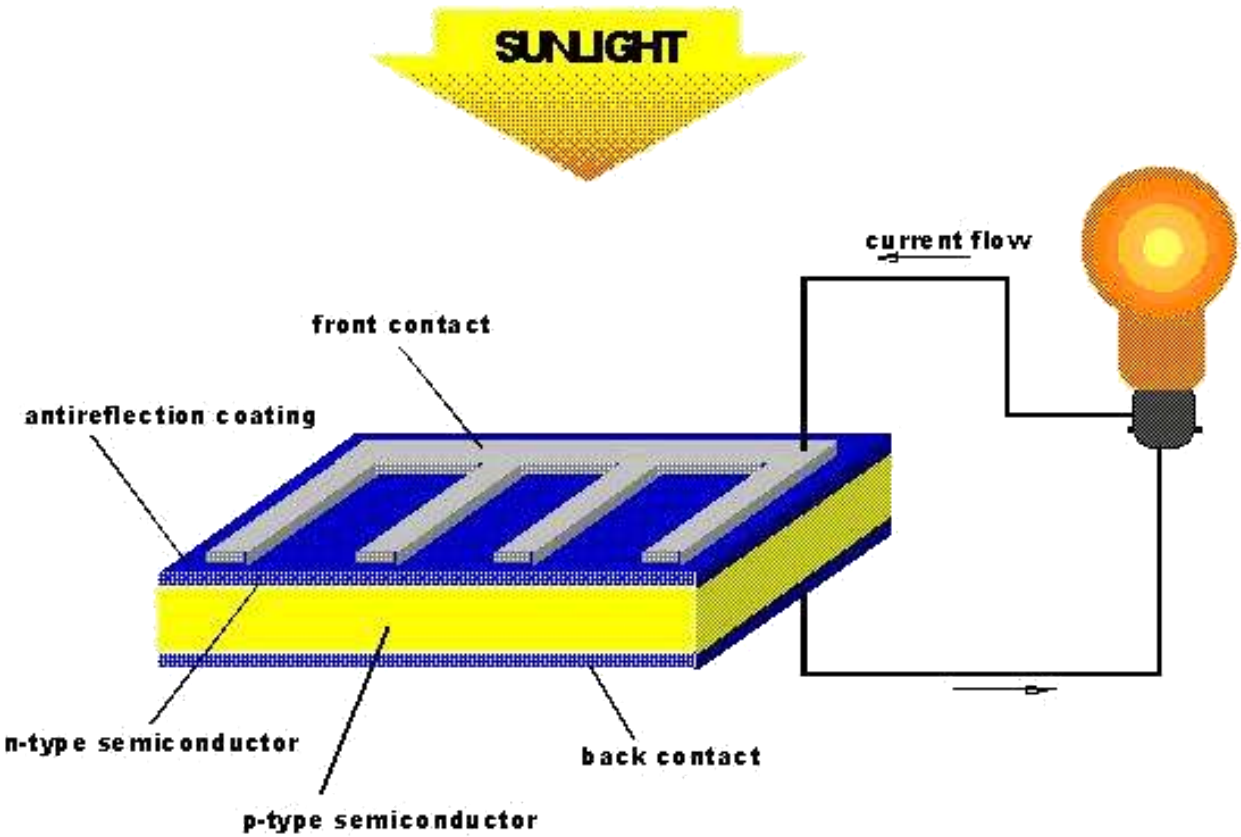


Figure 3: The Depletion Layer

http://www.americansolareconomy.com

PHOTOVOLTAIC CELL



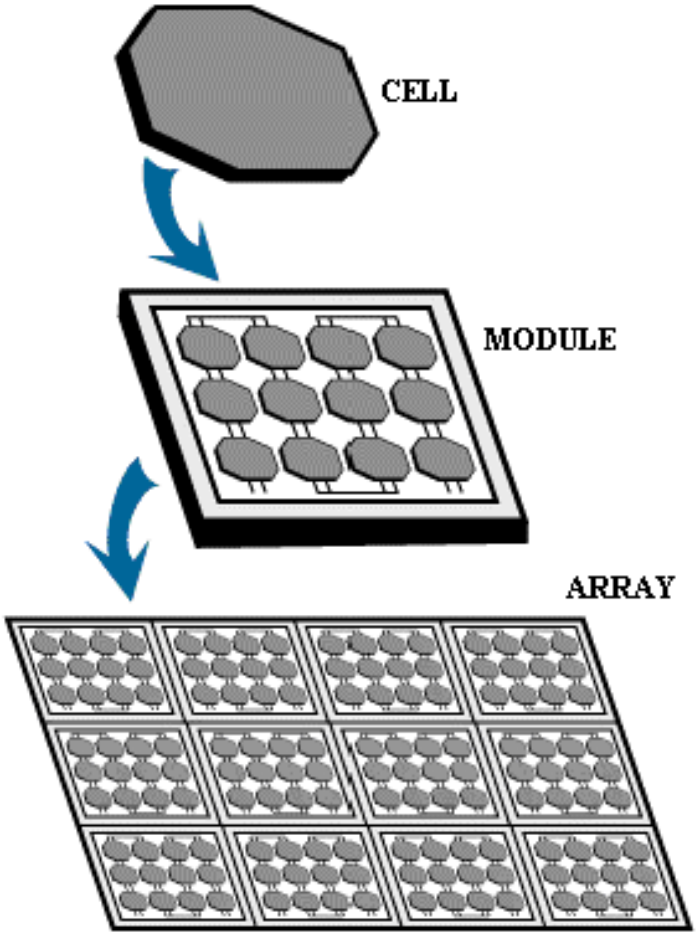
http://www.pv.unsw.edu.au/future-students/pv-devices/how-they-work.asp

PHOTOVOLTAIC SYSTEMS

- Crystalline silicon PV cells produces about 0.5 V electricity.
- The current depends on size and intensity of radiation.
- Typically 10x10 cm solar cell produces 2-5 A
- Hence typical power output is about 0.5 x(2-5)=1-2 W DC
- Then cells connected "in series" to increase voltage and parallel to increase "current" (usually 36 cells in series grouped together to form a modul, parallel connection is not often used)

PHOTOVOLTAIC SYSTEMS

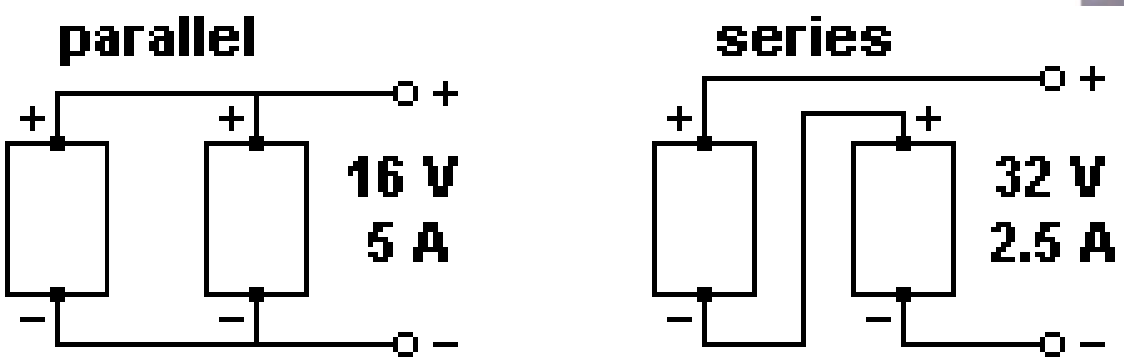
- The modules are combined into arrays (panels) to produce desired voltage and current (hence power)

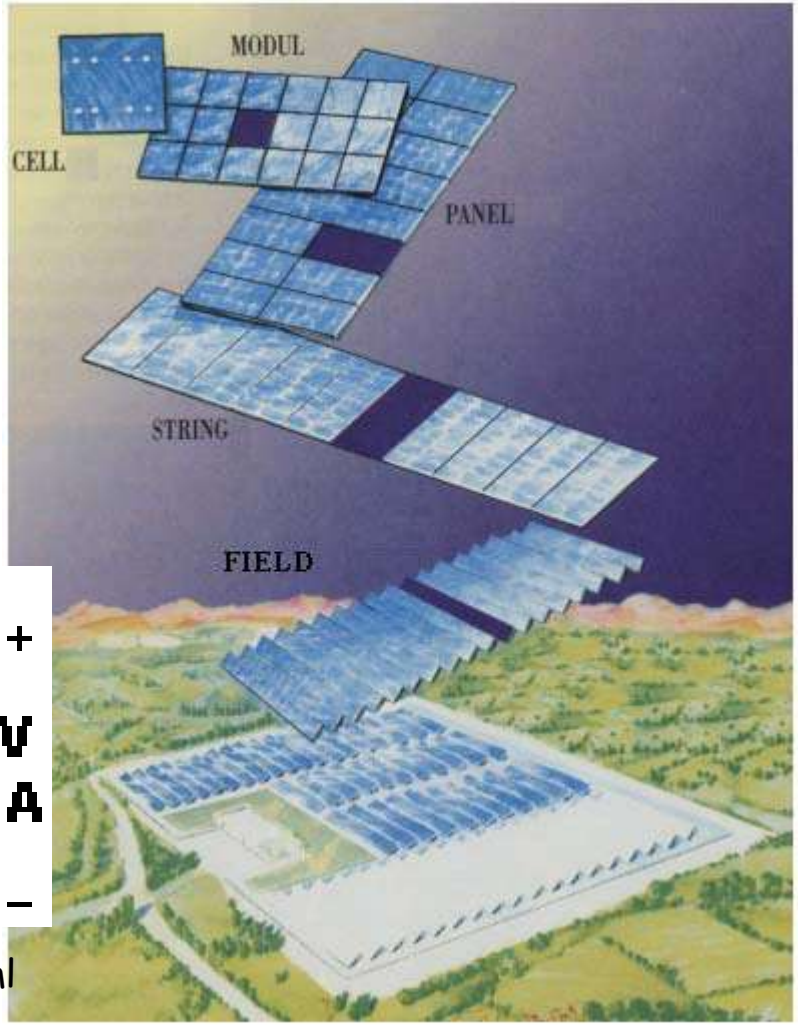


http://science.nasa.gov/headlines/y2002/solarcells.htm

PHOTOVOLTAIC SYSTEMS

- Arrays/panels combines to form solar fields
- Can have any voltage and/or current using parallel and series connection





http://www.jgsee.kmutt.ac.th/exell/Solar/PVSystems.html

PHOTOVOLTAIC SYSTEMS

- PV system consists of electrical, electronic and mechanical components
- System elements and configuration may vary:
  - Flat-plane vs. sun tracking systems (require electro/mechanical tracking devices)
  - Stand alone vs. grid connection (no battery needed)
  - Pure solar or hybrid systems (some additional power generator)
  - Size and application: utility power generation, home use, small remote applications such as water pumping, base stations, satellite .....
- But there are some common elements that exist in all systems

PHOTOVOLTAIC SYSTEMS

FLAT-PLANE SYSTEMS

- Built on fixed mechanical structures,
- They are generally tilted at an angle equal to the latitude to maximize electricity per cost
- Usually for low cost applications

http://www.e8.org/Projects/Majuro/Solar-PV_fichiers/3%20Photovoltaic%20Systems.pdf

PHOTOVOLTAIC SYSTEMS

SUN TRACKING SYSTEMS

- Tracking is used if the cost is not important compared to collecting the maximum energy.
- The plane of the module rotates according one axis ("single-axis") or two axes ("double-axis")
- Tracking needs actuation & control, which complicate the system.

http://www.e8.org/Projects/Majuro/Solar-PV_fichiers/3%20Photovoltaic%20Systems.pdf

PHOTOVOLTAIC SYSTEMS

SUN TRACKING SYSTEMS

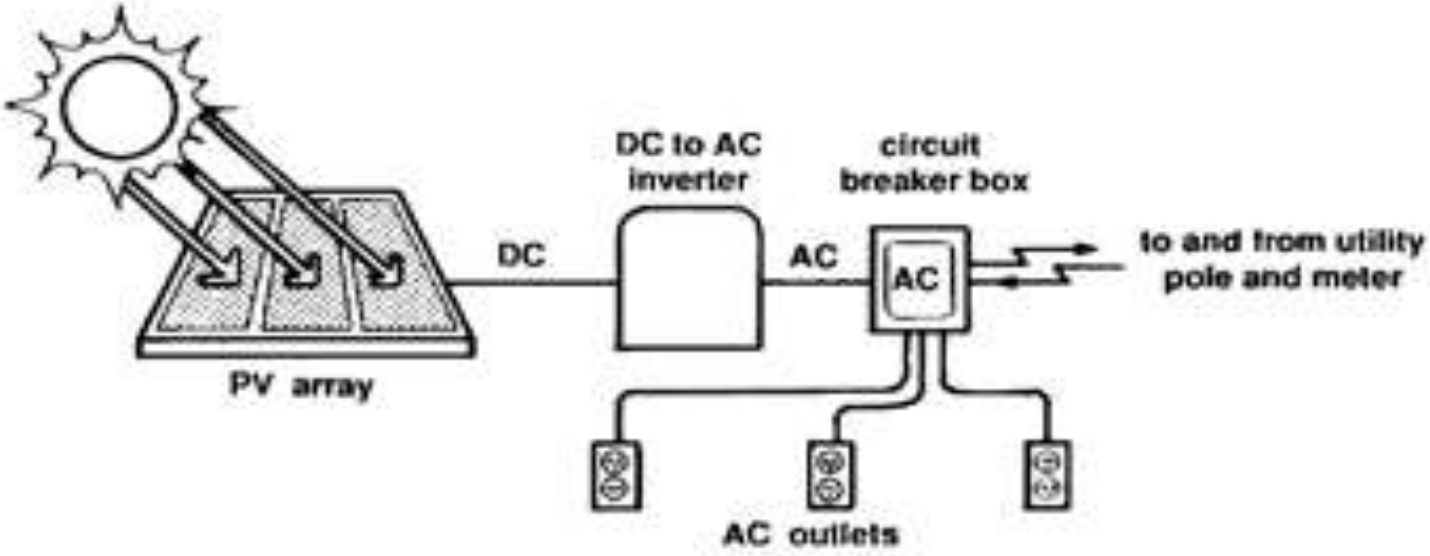
- It is possible to use "passive trackers"
- They are operated by exploiting the pressure differences that are generated in two mechanical pistons filled with special fluids, when one is exposed to and heated by the solar radiation while the second is in the shade.

http://www.e8.org/Projects/Majuro/Solar-PV_fichiers/3%20Photovoltaic%20Systems.pdf

PHOTOVOLTAIC SYSTEMS

Grid connected systems

- They are simpler because they do not require electric storage. PV produced electricity is given to the grid during the day time, while the grid supplies the electric energy at night.
- These systems are more reliable than a stand-alone set, which has to be oversized to cover all possible bad weather conditions.

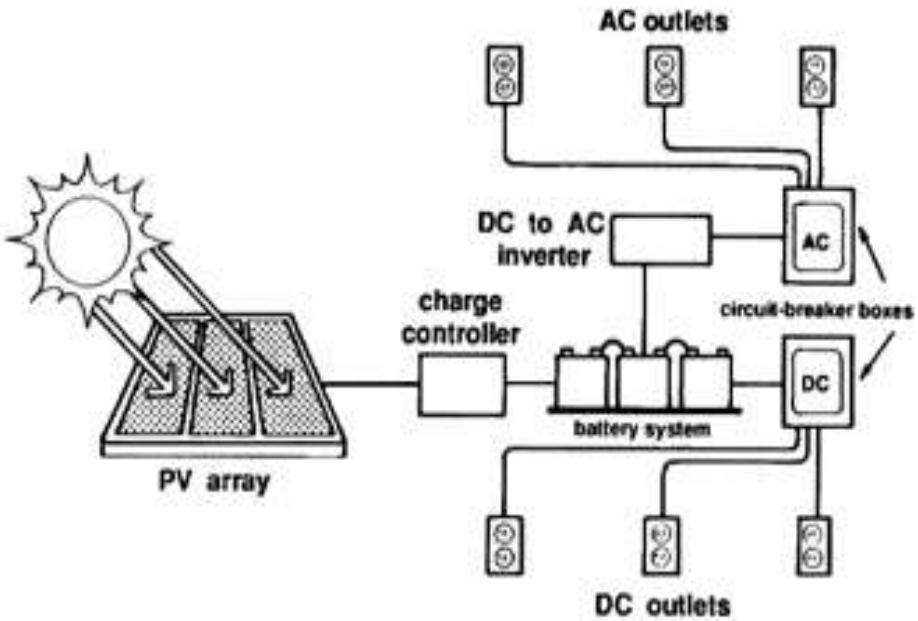


http://www.energy.ri.gov/programs/renewable.php

PHOTOVOLTAIC SYSTEMS

Stand-alone systems

- They are generally equipped with some form of storage to supply electricity when the solar radiation is not present or insufficient.
- They are common for the domestic usage or for lighting
- They are usually oversized



http://www.energy.ri.gov/programs/renewable.php

Battery

- It allows to satisfy peak loads without over-capacity, and supply energy in case of low/no radiation or faults.
- It should be
  - High efficiency
  - Long lifecycle in frequent charge/discharge regimes
  - High resistance at high thermal excursions ($0 \div 50^\circ C$)
  - Low auto-discharge
  - High ratio capacity/volume
  - Low cost
  - Low maintenance
- Usually, the lead-acid batteries are used in PV systems (benefiting from auto industry)

Charge Control

- It is necessary to control charge for lead-acid battery to keep the voltage below the limit of 2.4 Volt/element to avoiding gas production.
- A charging control cuts off the battery charging current when the voltage is greater than a maximum

PHOTOVOLTAIC SYSTEMS

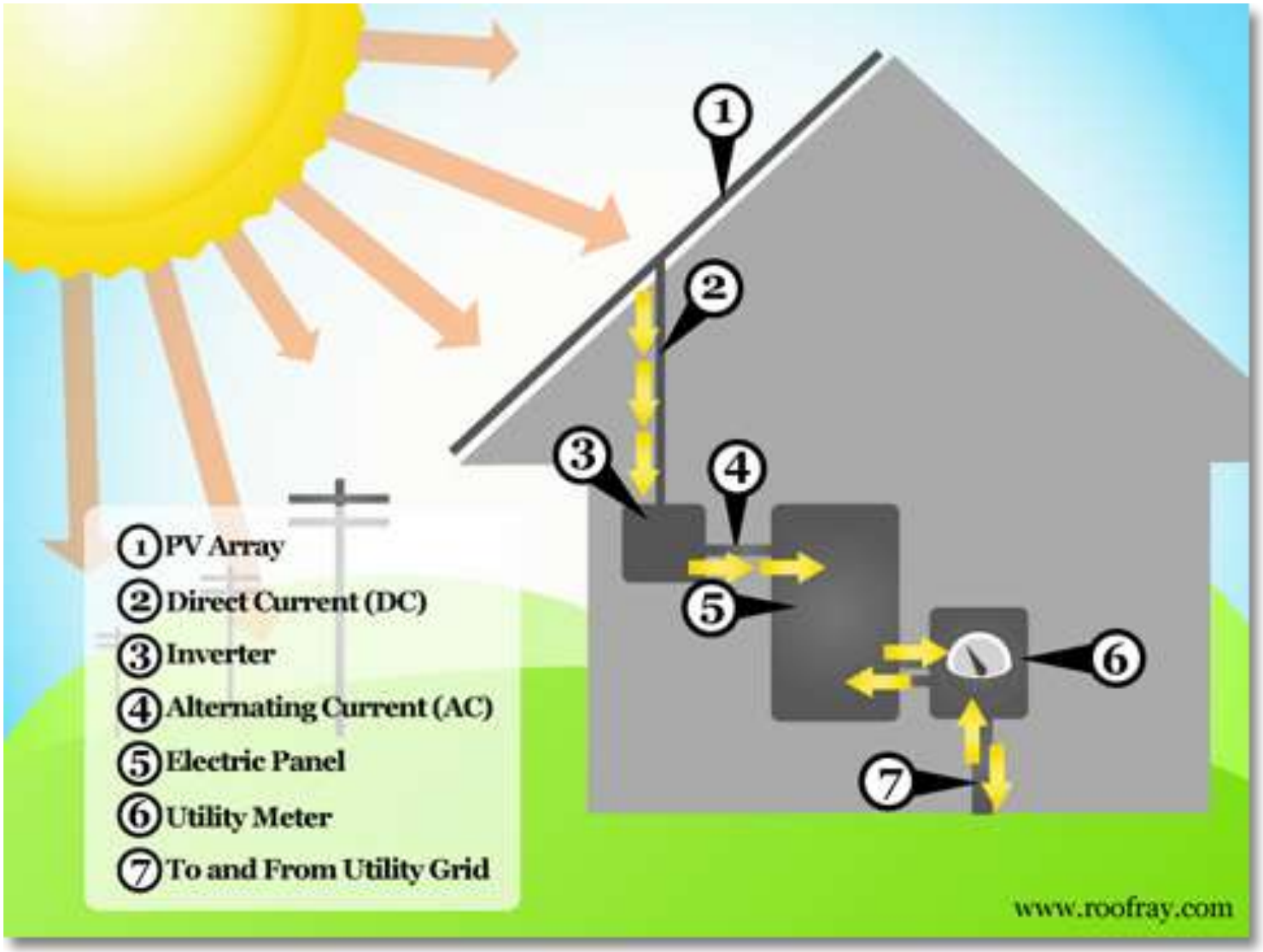
DC/AC Convertor

- A DC/AC convertor is needed to convert the DC current produced and stored in the system for AC powered applications.
- DC /AC convertor should have the following properties:
  - High DC/AC conversion rate;
  - Low consumption without load;
  - Low frequency distortion;
  - Reduced weight and sizes;
  - High reliability.

Other Elements

- Power distribution panels, metering devices, fuses, by-pass and blocking diodes....

PHOTOVOLTAIC SYSTEMS



PHOTOVOLTAIC CELL MATERIAL

- Silicon technologies
  - Single crystal silicon (c-Si)
  - Multicrystalline silicon (mc-Si)
  - Amorphous silicon (a-Si)
- Polycrystalline thin film technologies
  - Cadmium telluride (CdTe)
  - Copper indium gallium diselenide (CIGS) alloy
  - Thin film silicon (Nanocrystalline, nc-Si or small-grained polycrystalline silicon,pc-Si)
- Single-crystal Gallium Arsenide (GaAs)
- Organic/polymeric solar cells
  - Double layer, bulk heterojunction,
  - Dye sensitized
  - Organolead halide perovskites

PHOTOVOLTAIC CELL MATERIAL

The basic criteria that can be used to compare the materials are: Crystallinity, adsorption, bandgap, cost and complexity of production….

Crystallinity: indicates how perfectly ordered the atoms are in the crystal structure. One way of classification wrt crystallinity is approximate crystal size

Type of Silicon	Abbreviation	Crystal Size Range	Deposition Method
Single-crystal silicon	sc-Si	>10cm	Czochralski, float zone
Multicrystalline silicon	mc-Si	1mm-10cm	Cast, sheet, ribbon
Polycrystalline silicon	pc-Si	1mm	Chemical-vapor deposition
Microcrystalline silicon	mc-Si	<1mm	Plasma deposition

http://www1.eere.energy.gov/solar/solar_cell_materials.html

PHOTOVOLTAIC CELL MATERIAL

Absorption:

- The absorption coefficient indicates how far light having a specific wavelength (or energy) can penetrate the material before being absorbed.
- A small absorption coefficient means that light is not readily absorbed by the material.
- Depends on the material making and the wavelength or energy of the light being absorbed.

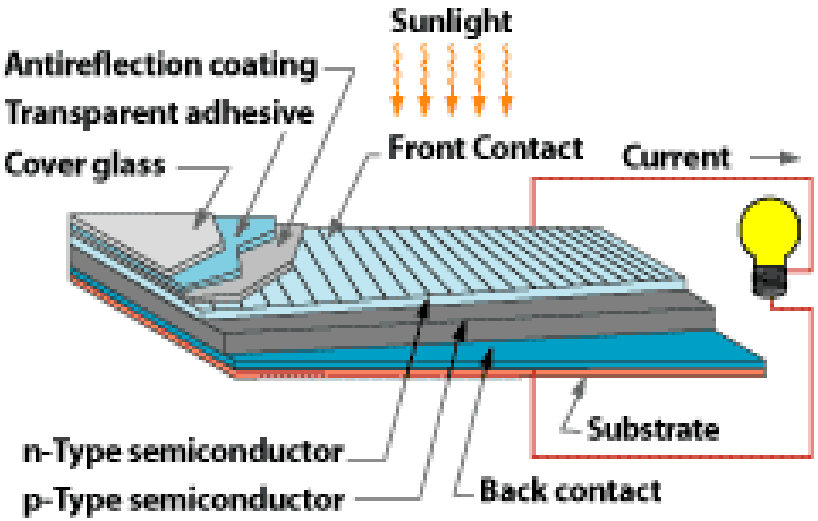
http://www1.eere.energy.gov/solar/solar_cell_materials.html

PHOTOVOLTAIC CELL MATERIAL

Bandgap : minimum energy needed to move an electron from its bound state (valance) within an atom to a free state (conduction).

Cost and complexity of production:

- Production of material and cell
- Amount and type of material, number of steps involved....
- Semiconductor layer is the most important part



http://www1.eere.energy.gov/solar/solar_cell_materials.html

SINGLE CRISTAL SILICON

Silicon

- The first and still the most common material used in semiconductor production for electronics and PV devices
- The second-most abundant element in the Earth's crust after oxygen.
- It has to be very pure (99.9999%) to be used as semiconductor in electronics, could be less pure for PV.

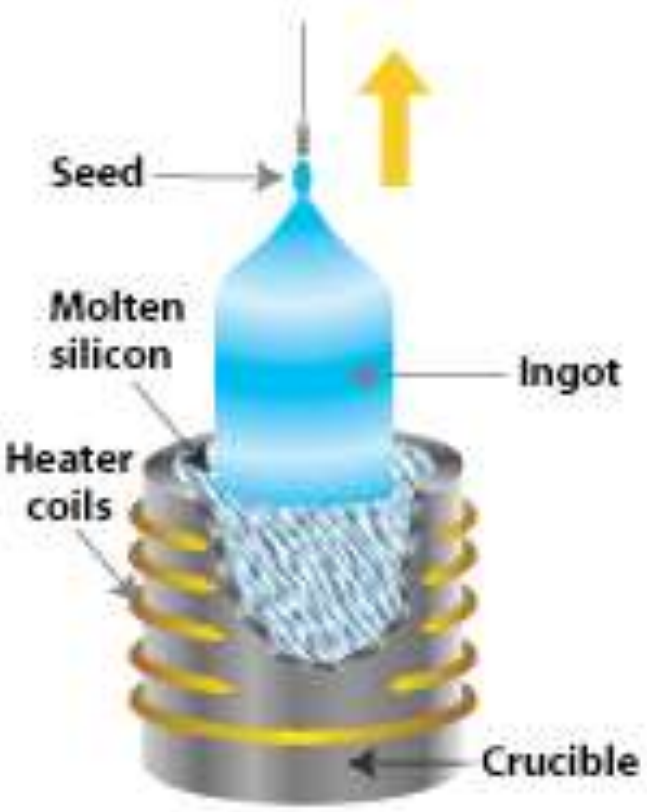
Single-crystal silicon

- The entire structure is grown from the same crystal
- The best silicon for the solar cells but it is expensive
- Various techniques are used for production

SINGLE CRISTAL SILICON

The Czochralski Process

- High-purity Si melted down in a crucible first
- A seed crystal is dipped into a crucible of molten silicon and withdrawn slowly, pulling a cylindrical single crystal as the silicon crystallizes on the seed.
- The impurities are generally cumulated in the inferior part of the ingot.



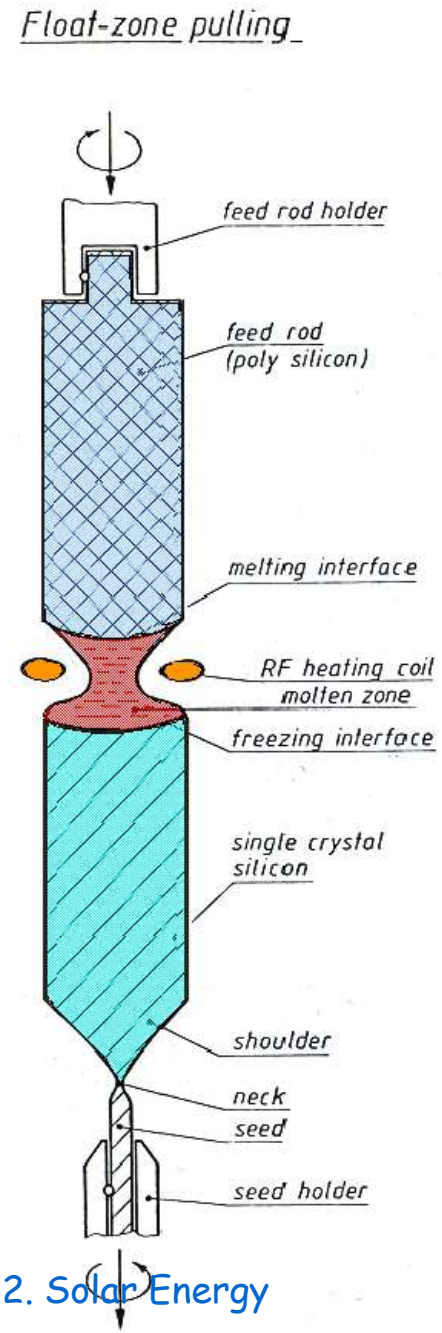
http://www1.eere.energy.gov/solar/printable_versions/silicon.html

SINGLE CRISTAL SILICON

Float-Zone Silicon

- A silicon rod is set atop a seed crystal and then lowered through an electromagnetic coil.
- The coil's magnetic field induces an electric field in the rod, heating and melting the interface between the rod and the seed. Single-crystal silicon forms at the interface, growing upward as the coils are slowly raised.
- It produces purer crystals than the Czochralski method,

http://www.tf.uni-kiel.de/matwis/amat/elmat_en/kap_6/advanced/t6_1_3.html



SINGLE CRISTAL SIL

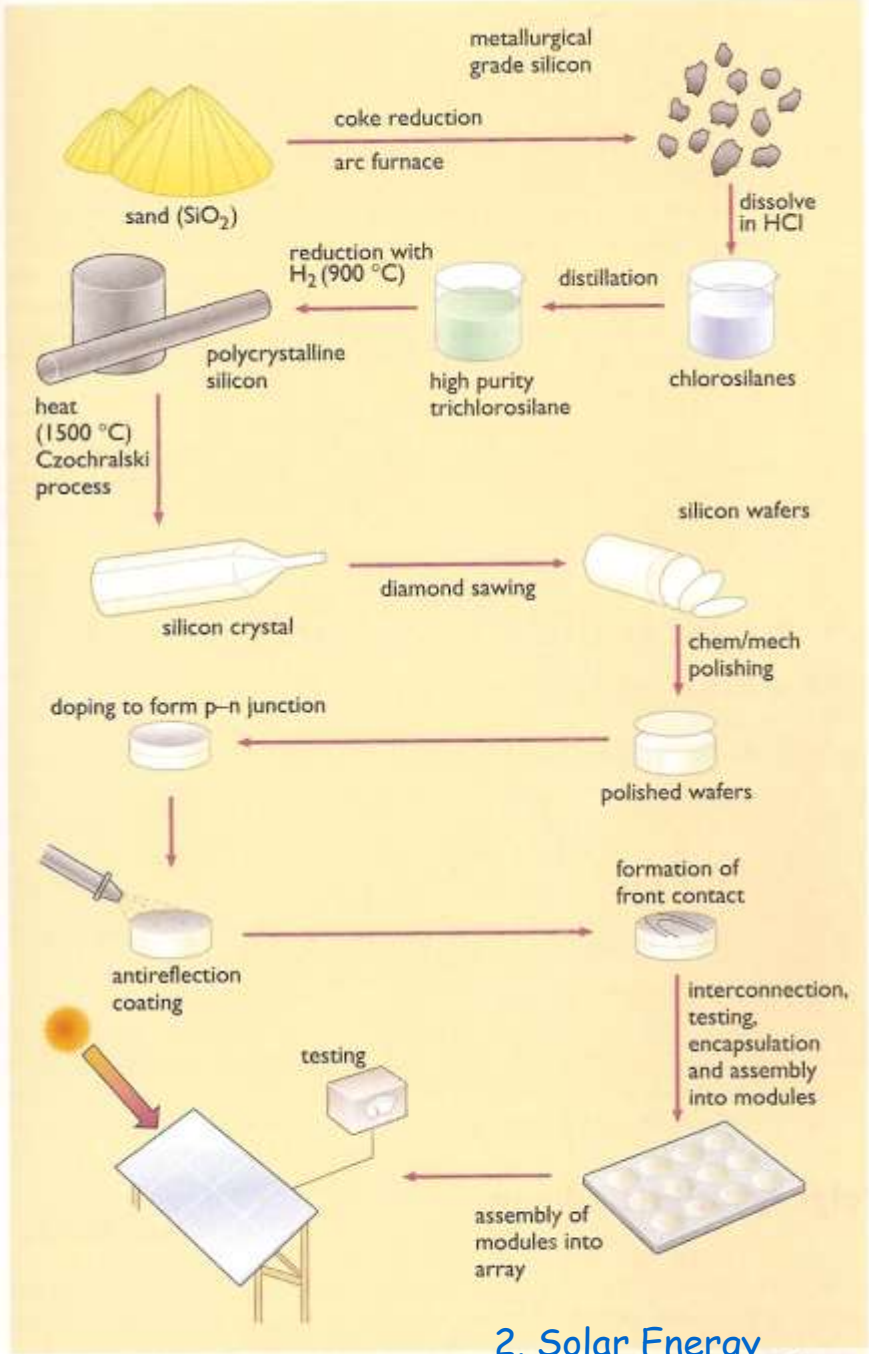
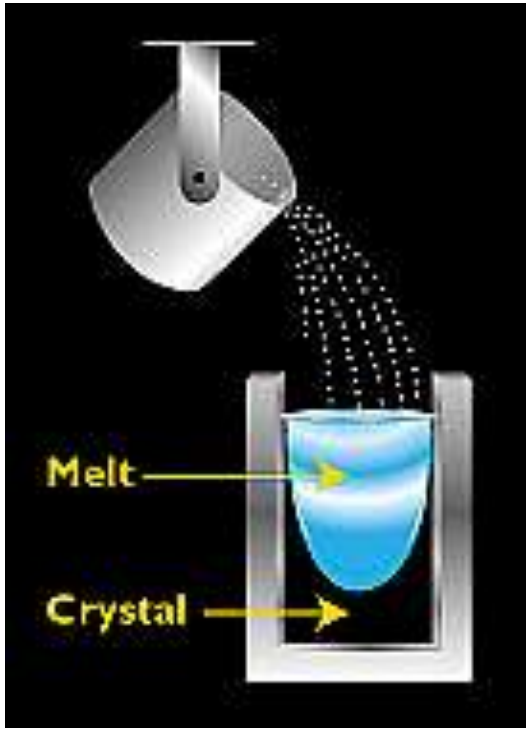


Figure 3.12 The overall process of monocrystalline silicon solar cell and module production

Boyle, Renewable Energy, Oxford,2004

MULTICRISTALLINE SILICON

- It is less efficient than single-crystal silicon, but also less expensive
- The most popular commercial methods involve casting molten silicon directly into a mold and allowed to solidify into an ingot.
- Lower grade silicon can be used in contrast to single-crystal material that require high grade.
- The mold is usually square producing ingot that can be cut into square cells fitting more compactly into a PV module.



http://www1.eere.energy.gov/solar/printable_versions/silicon.html

AMORPHOUS SILICON

- Atoms of amorphous silicon are not arranged in particular order as in crystals.
- It absorbs solar radiation 40 times more efficiently than single-crystal silicon, about 1 micrometer film can absorb 90% of the usable light . That's reduces the cost of cell significantly
- It can also be produced at lower temperatures and can be deposited on low-cost substrates such as plastic, glass and metal
- However, the efficiency is low and it decreases with time (about 20%) until it stabilizes.

AMORPHOUS SILICON

- The structure of amorphous slicon cells are sligthtly different than the cristalline cells

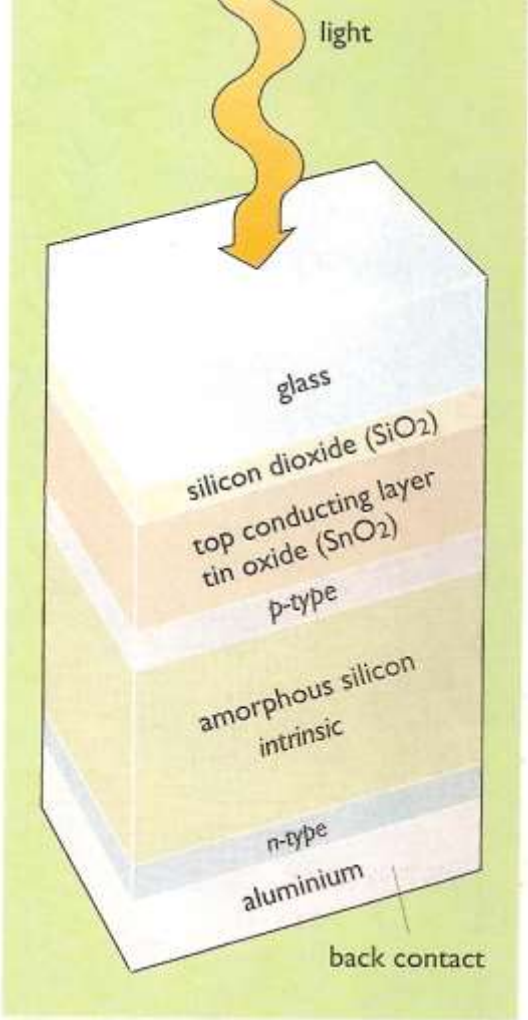


Figure 3.17 Structure of an amorphous silicon cell. The top electrical contact is made of an electrically conducting, but transparent, layer of tin oxide deposited on the glass. Silicon dioxide forms a thin 'barrier layer' between the glass and the tin oxide. The bottom contact is made of aluminium. In between are layers of p-type, intrinsic and n-type amorphous silicon

Boyle, Renewable Energy, Oxford,2004

THIN FILM TECHNOLOGIES

- "Thin film" comes from method to deposit film, not from thinness of film): Cells are deposited in thin, consecutive layers of atoms, molecules, or ions.
- These methods use much less material—the cell's active area is usually only 1 to 10 micrometers thick, whereas thick films typically are 100 to 300 micrometers.
- Can be manufactured in large areas
- They can be deposited on any surface such as steel, glass flexible plastics....
- Several deposition techniques can be used (physical vapor deposition, chemical vapor deposition, electrochemical deposition, or a combination.
- All are potentially less expensive than crystalline silicon

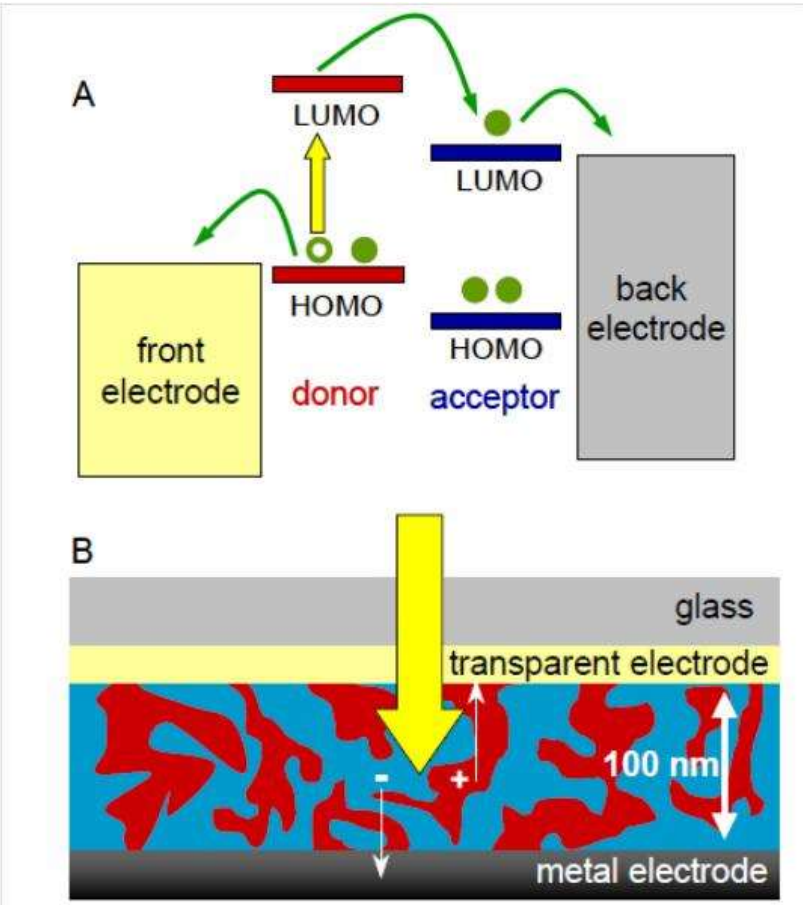
THIN FILM TECHNOLOGIES

http://www.guardian.co.uk/environment/gallery/2009/may/21/copenhagen-climate-change-manchester-report?picture=347729522

GALLIUM ARSINIDE

- Gallium arsenide (GaAs) is a compound semiconductor: a mixture of gallium and arsenic.
- It is used in the form of single crystal
- It is also used in LEDs, lasers & other electronic devices
- Advantages:
  - Its band gap of 1.43 eV is nearly ideal
  - High absorptivity: Only a few microns layer is sufficient compared to Si that requires 100 micron or more
  - Relatively insensitive to heat, resistant to radiation damages
  - It has design flexibility
  - Can be alloyed with Al, P, Pb, Ir for more flexibility
- But it is expensive, feasible in highly efficient systems like concentrators

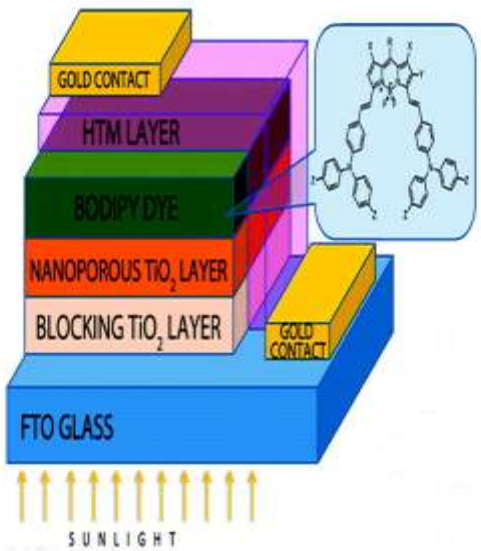
ORGANIC SOLAR CELLS



A: double layer B: Bulk heterojunction

- Front electrode (Cathode): ITO or some organic material
- Donor: a conjugated polymer
- Acceptor: a fullerene derivative
- Back electrode (Anode): Usually Al
- Cathode material is coated on a glass (or transparent polymer)
- Donor and acceptor are coated over the cathode sequentially (in double layer) or as mixture (in bulk heterojuntion)
- Metal anode coated over the donar/acceptor material
- Techniques used: Sping coating, ink-jet printing, roll-to-roll, chemical/ physical vapor deposition….

DYE/ORGANOLEAD SENSITIZE SOLAR CELLS

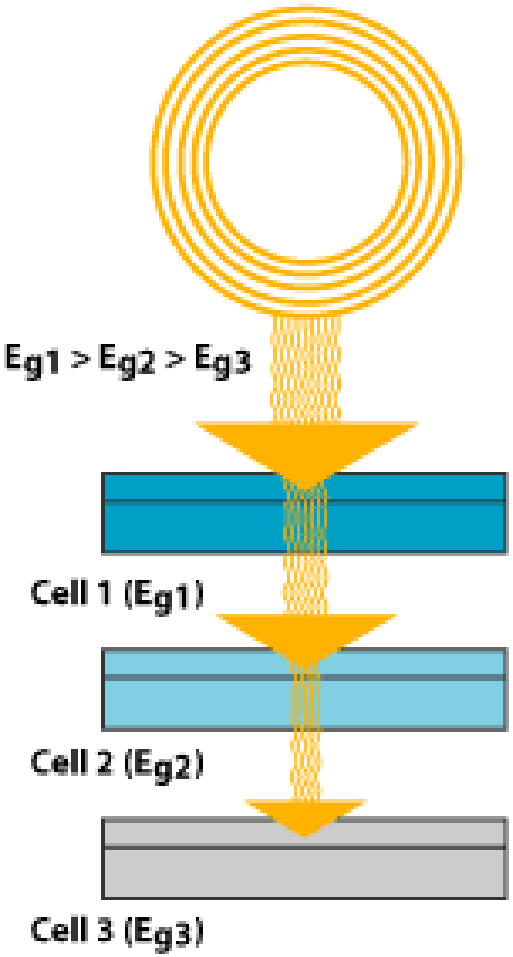


- Imitates photosynthesis
- A light sensitive dye absorb the light and get excited
- Transfer its electron to a semiconductor
- The electron transferred to back electrode using a electrolyte solution (or solid state material)
- Semiconductor is usually TiO2
- Ru based synthetic dyes or natural dyes from fruits and vegetables are used
- Organolead halide perovskites are used as a new and more effective sensitizers

BASIC CELL STRUCTURE

There are four basic structure for pV cells

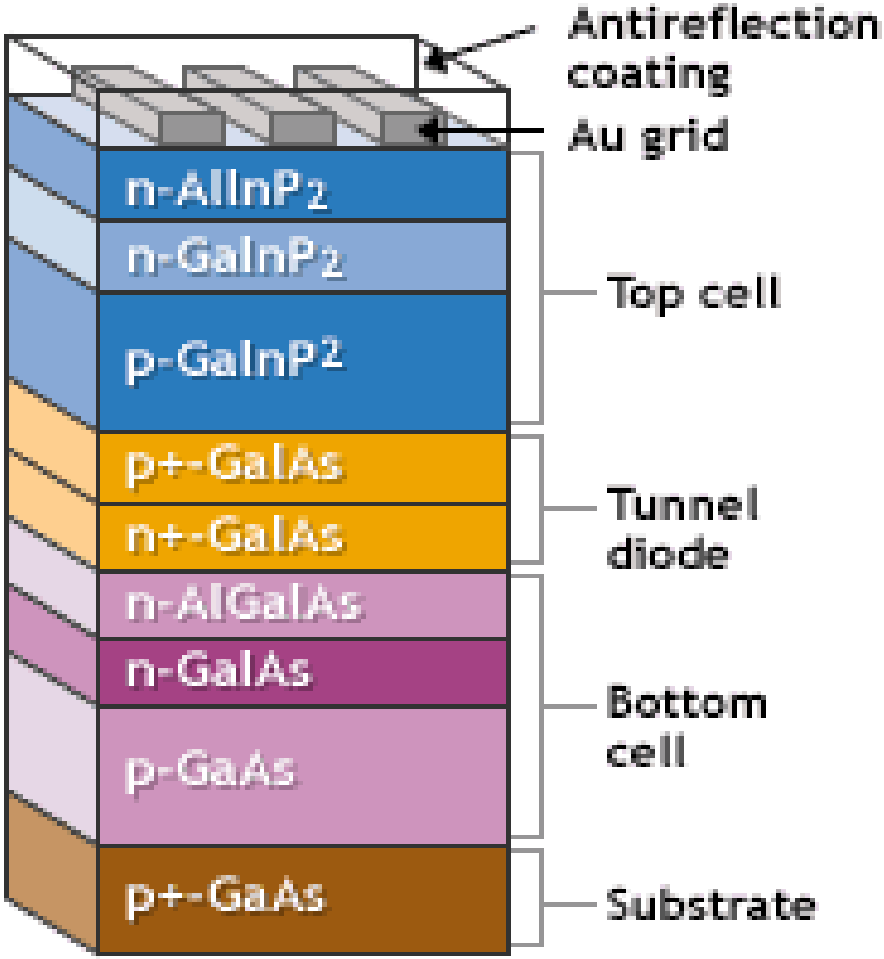
- Homojunction: As in single cristal Si
- Heterojunction: junction is formed by contacting two different semiconductor. (Ex: copper indium diselenide , cadmium telluride)
- p-i-n/n-i-p: As in amorphous Si
- Multijunction: Individual cells with different bandgaps (decreasing down) are stacked on top of one another for higher total conversion efficiency



http://www1.eere.energy.gov/solar/bandgap_energies.html

BASIC CELL STRUCTURE

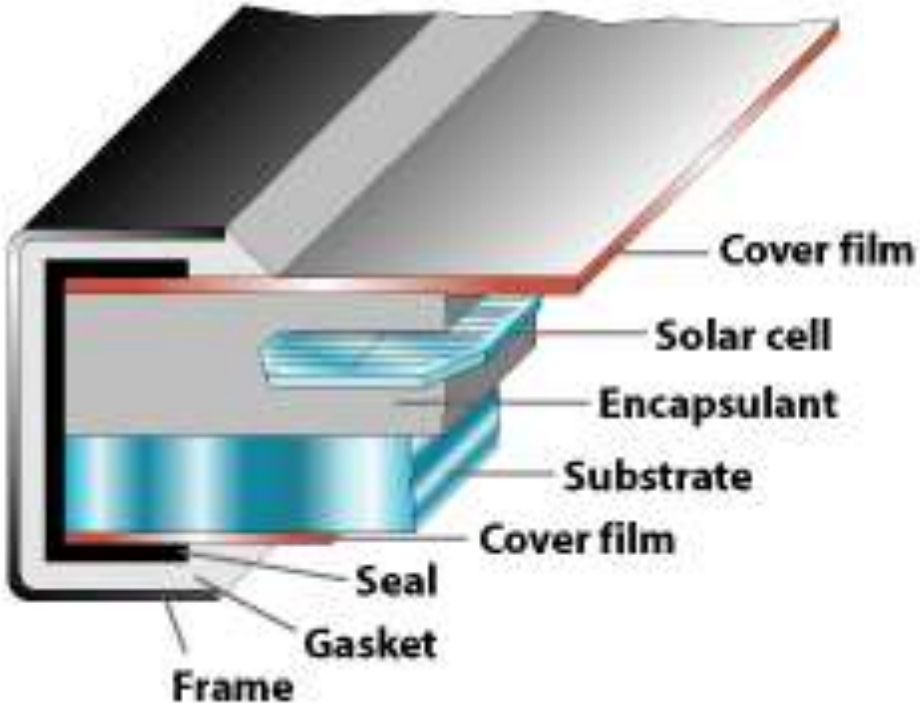
Multijunction cell



http://www1.eere.energy.gov/solar/bandgap_energies.html

PANEL STRUCTURE

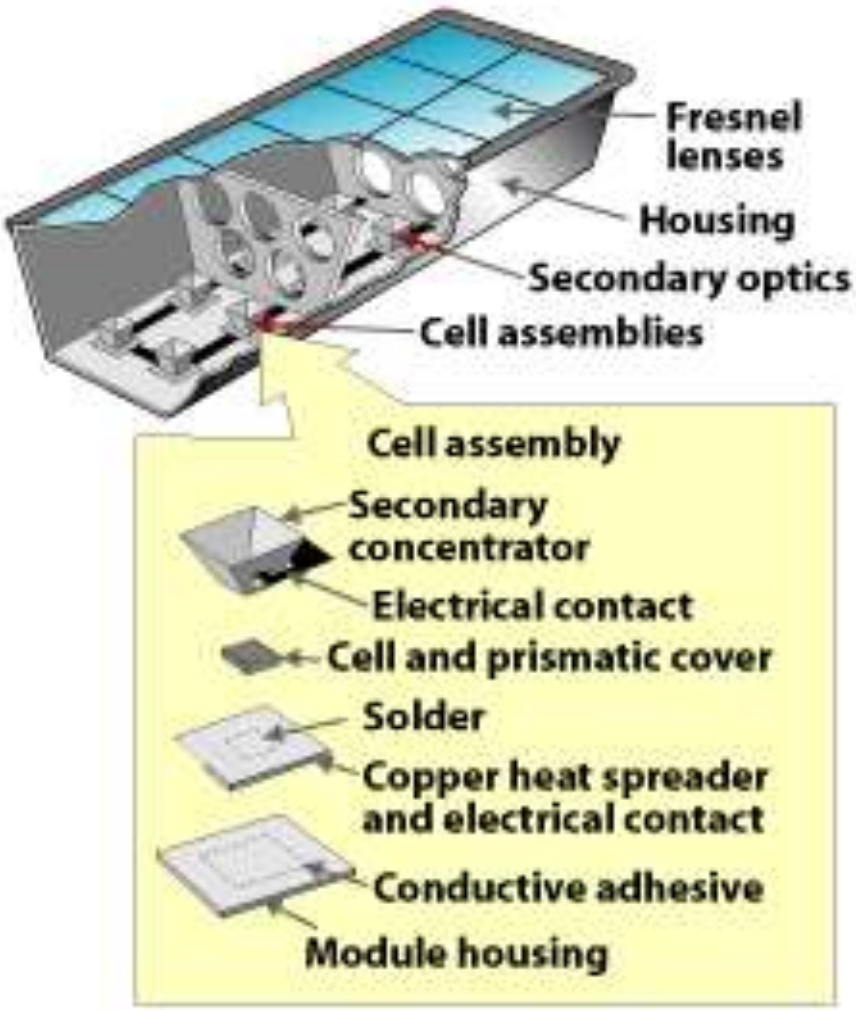
Typical flat-plate module design uses a substrate of metal, glass, or plastic to provide structural support in the back; an encapsulant material to protect the cells; and a transparent cover of plastic or glass.



http://www1.eere.energy.gov/solar/bandgap_energies.html

PANEL STRUCTURE

- A typical basic concentrator unit consists of a lens to focus the light, cell assembly, housing element, secondary concentrator, mechanism to dissipate excess heat produced by concentrated sunlight, and various contacts and adhesives..



http://www1.eere.energy.gov/solar/bandgap_energies.html

PERFORMANCE AND EFFICIENCY

Current vs. Voltage

- Short-circuit current (I_{sc}): the current produced when the positive and negative terminals of the cell are short-circuited (V=0)
- Open-circuit voltage (V_{oc}) is the voltage across the positive and negative terminals under open-circuit conditions (I=0).

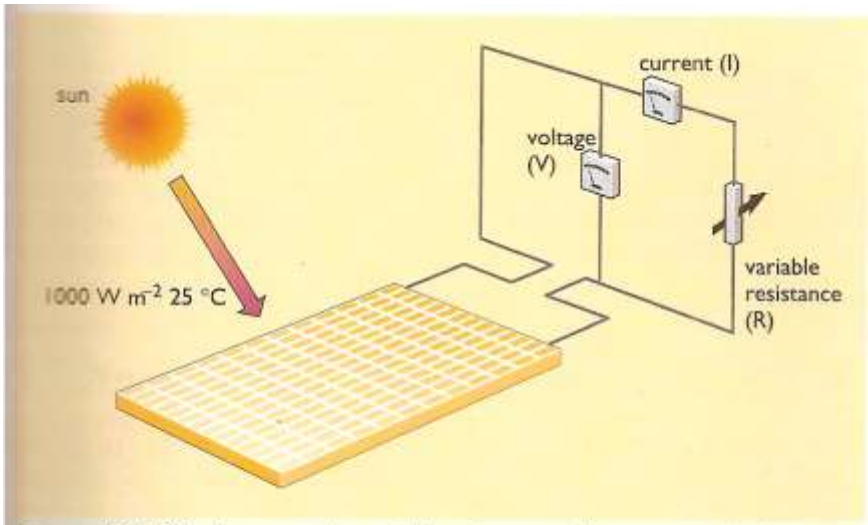


Figure 3.21 PV cell connected to variable resistance, with ammeter and voltmeter to measure variations in current and voltage as resistance varies

Boyle, Renewable Energy, Oxford, 2004

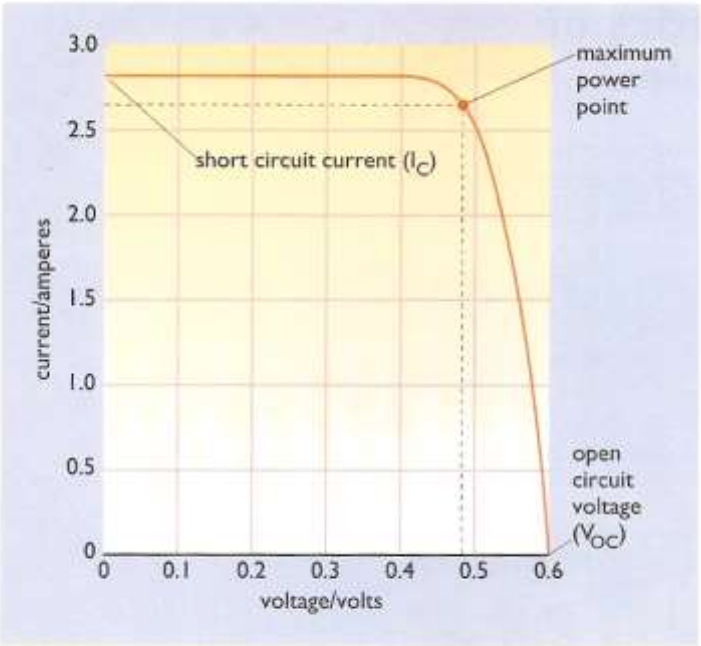
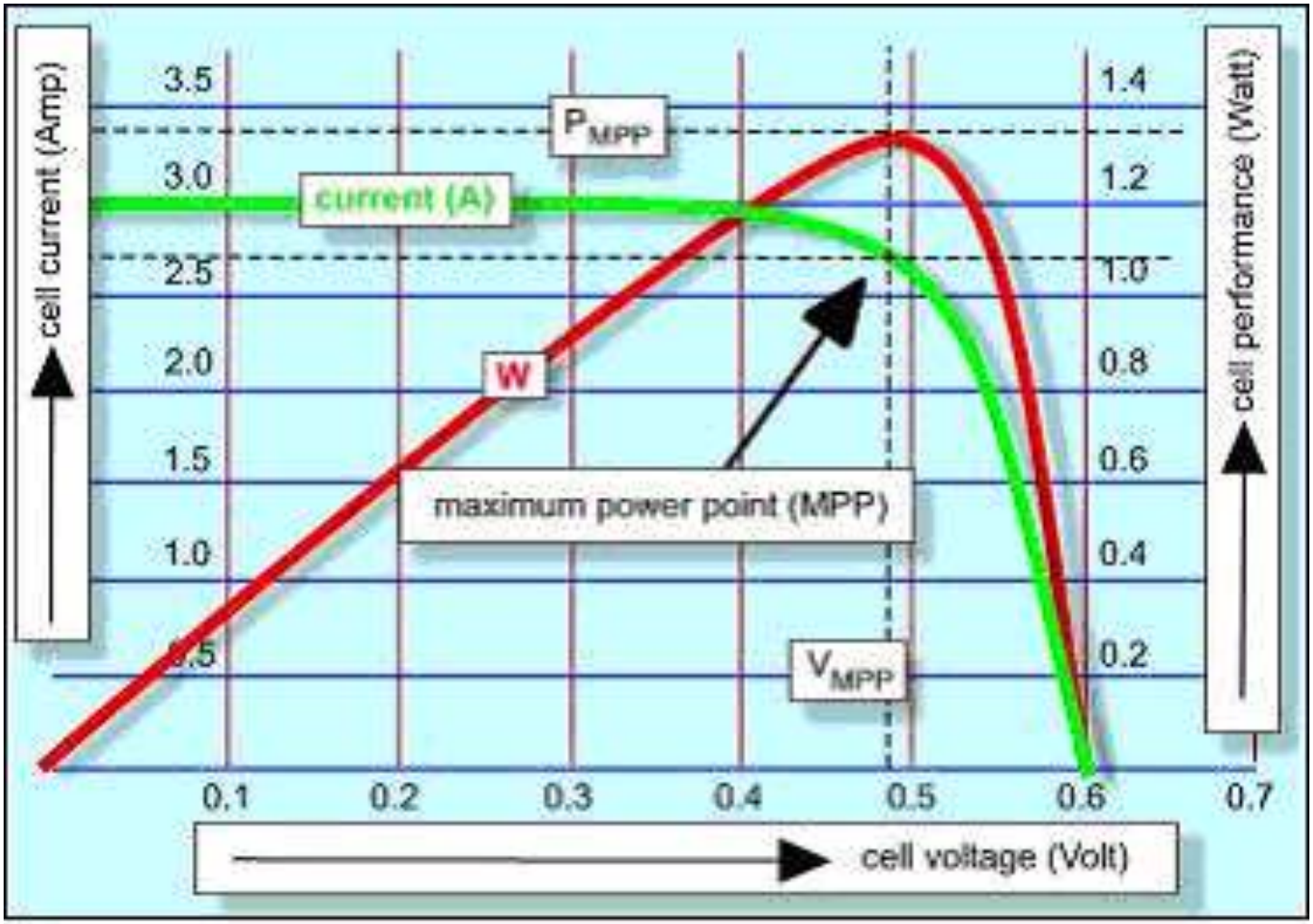


Figure 3.22 Current–voltage (I–V) characteristics of a typical silicon PV cell under standard test conditions

EFFICIENCY AND PERFORMANCE

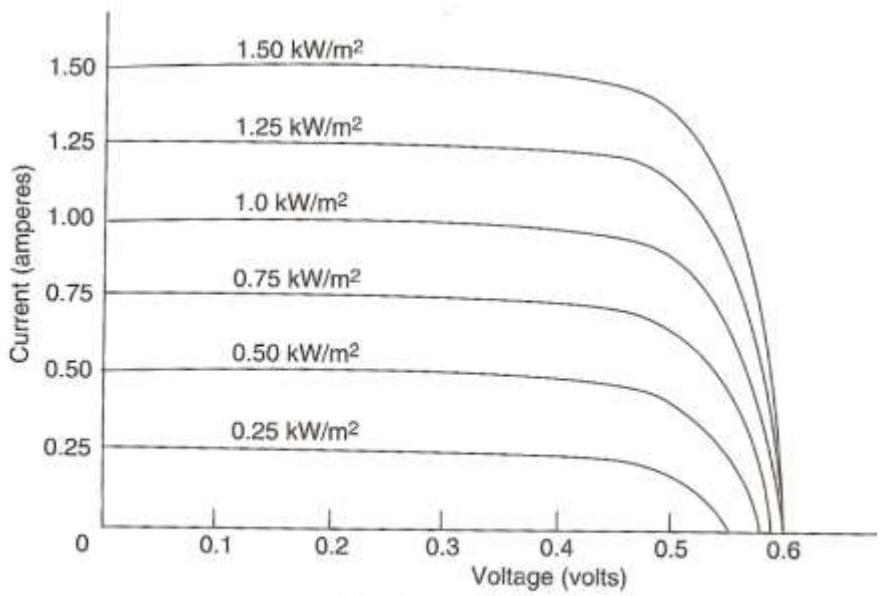


P=IxV

http://www.engineering.com/SustainableEngineering/RenewableEnergyEngineering/SolarEnergyEngineering/Photovoltaics/tabid/3890/Default.aspx

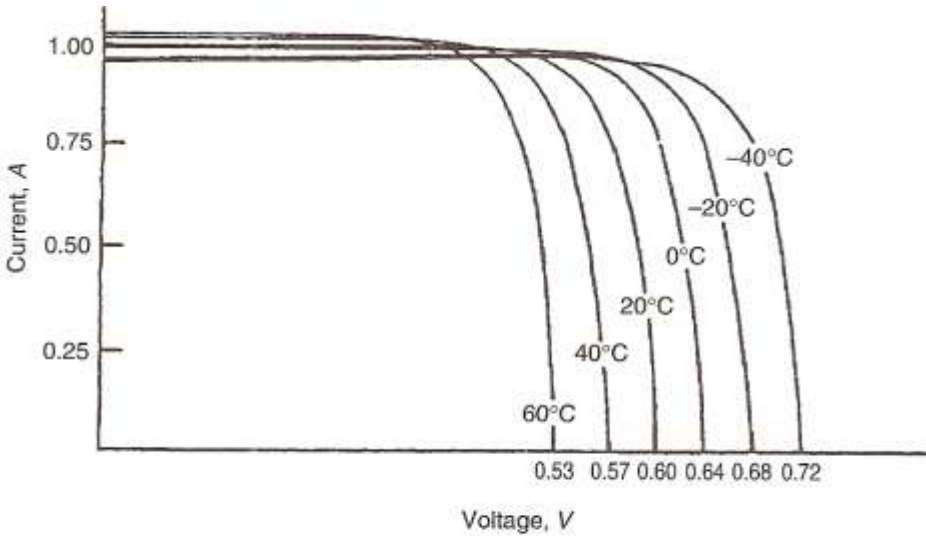
EFFICIENCY AND PERFORMANCE

- Current depends on "insolation" and"area of cell"
- Current is decreased with decreasing insolation (voltage does not change much)
- Voltage is decreases with increasing temperature



Changing current with changing insolation

(b)



Changing voltage with changing temperature

(c)

Figure 24 (*Continued*)

Kutz, Alternative Energy Production, Wiley, 2007

EFFICIENCY AND PERFORMANCE

Fill factor: measures the "squareness" of the I-V curve and describes the degree to which the voltage at the maximum power point (V_{mp}) matches V_{oc} and that the current at the maximum power point (I_{mp}) matches I_{sc}.

$$FF=(V_{mp}I_{mp})/(V_{oc}*I_{sc})$$

The higher the fill factor's percentage or match, the "squarer" the curve and better the cell

EFFICIENCY AND PERFORMANCE

Conversion efficiency: Percentage of the solar energy shining on a PV device that is converted into electrical energy

$$\eta = P_{mp}/(E*A)$$

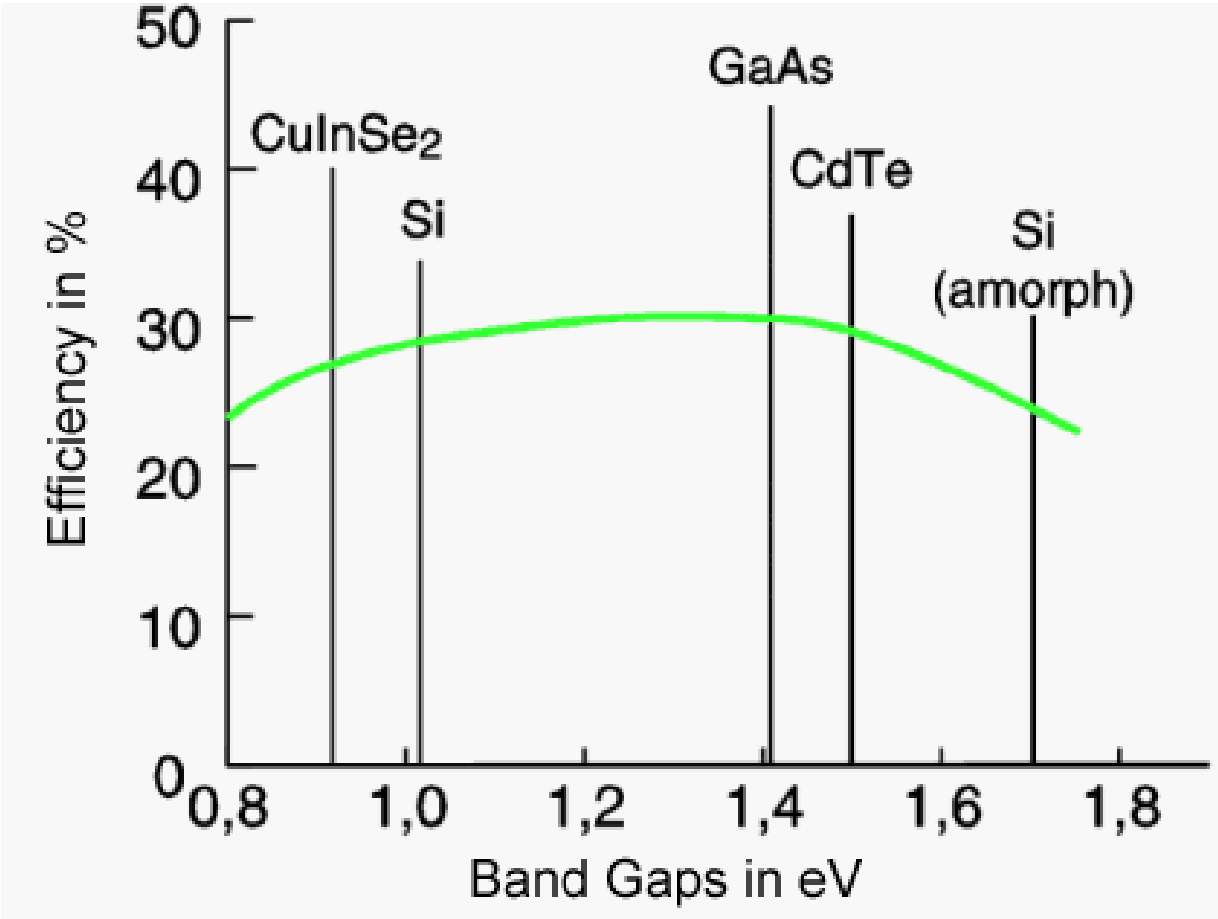
P_{mp}: maximum power can be produced, E: Insolation, A area of the cell.

η is usually calculated at E=1000 W/m^2, AM=1.5 and T=25 ^oC (standard test conditions)

EFFICIENCY AND PERFORMANCE

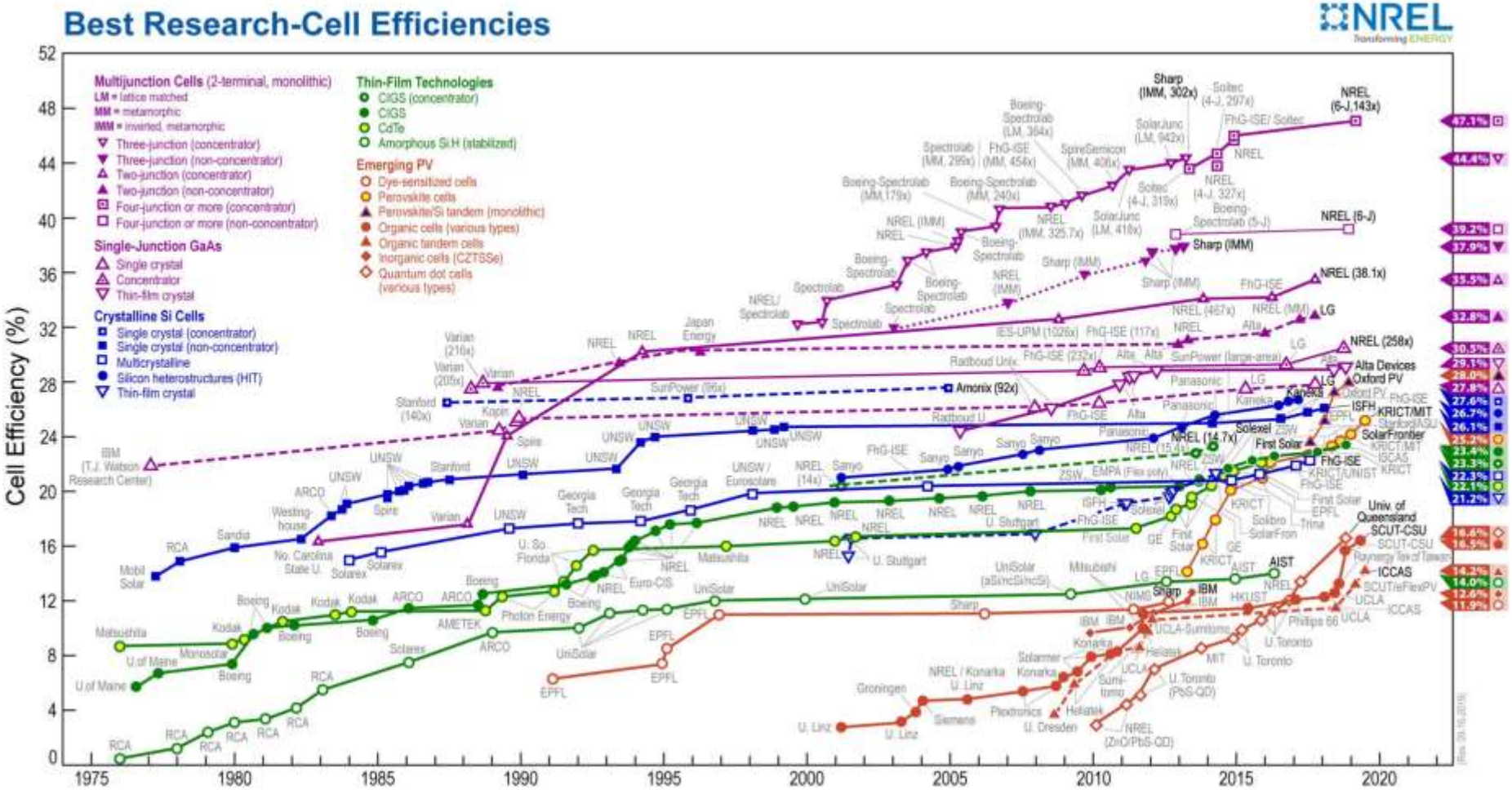
- Only the photons that have energy higher than bandgap energy of cell material can be converted to electricity.

- This creates a theoretical limits for efficiency depending on the material type.



http://www.solarserver.de/wissen/photovoltaik-e.html

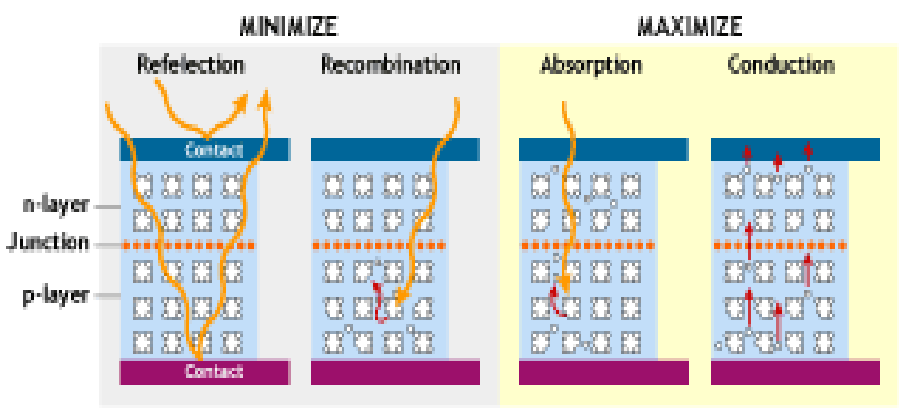
EFFICIENCY AND PERFORMANCE



EFFICIENCY AND PERFORMANCE

There are more sources of inefficiencies

- Recombination
- Blocking the light by metal contact
- Reflection of the light
- Impurities
- Temperature



http://www1.eere.energy.gov/solar/absorption_conduction.html

Prevention of reflection:

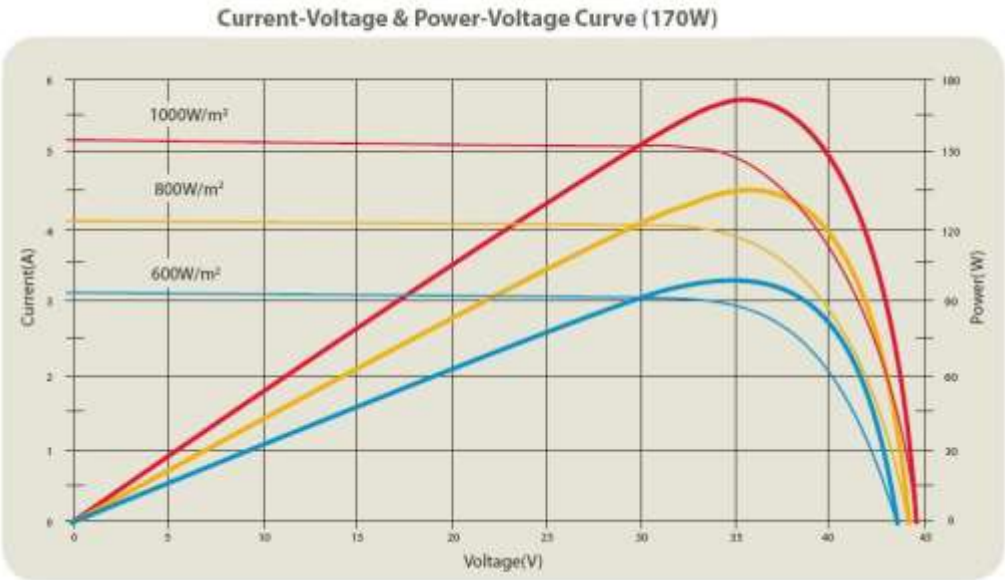
- Silicon is a shiny gray material reflecting more than 30% of the light
- Two techniques are commonly used to reduce reflection.
  - Coating with an antireflective material (like SiO)
  - Texturing the surface by chemical etching and creating pattern of cones and pyramids, which capture light rays

EFFICIENCY AND PERFORMANCE

Open - Circuit Voltage (Voc)	44.4V
Optimum Operating Voltage (Vmp)	35.6V
Short - Circuit Current (Isc)	5.40A
Optimum Operating Current (Imp)	5.05A
Maximum Power at STC (Pmax)	180Wp
Operating Temperature	-40ºC to +85ºC
Maximum System Voltage	1000V DC
Maximum Series Fuse Rating	10A
Power Tolerance	±3 %

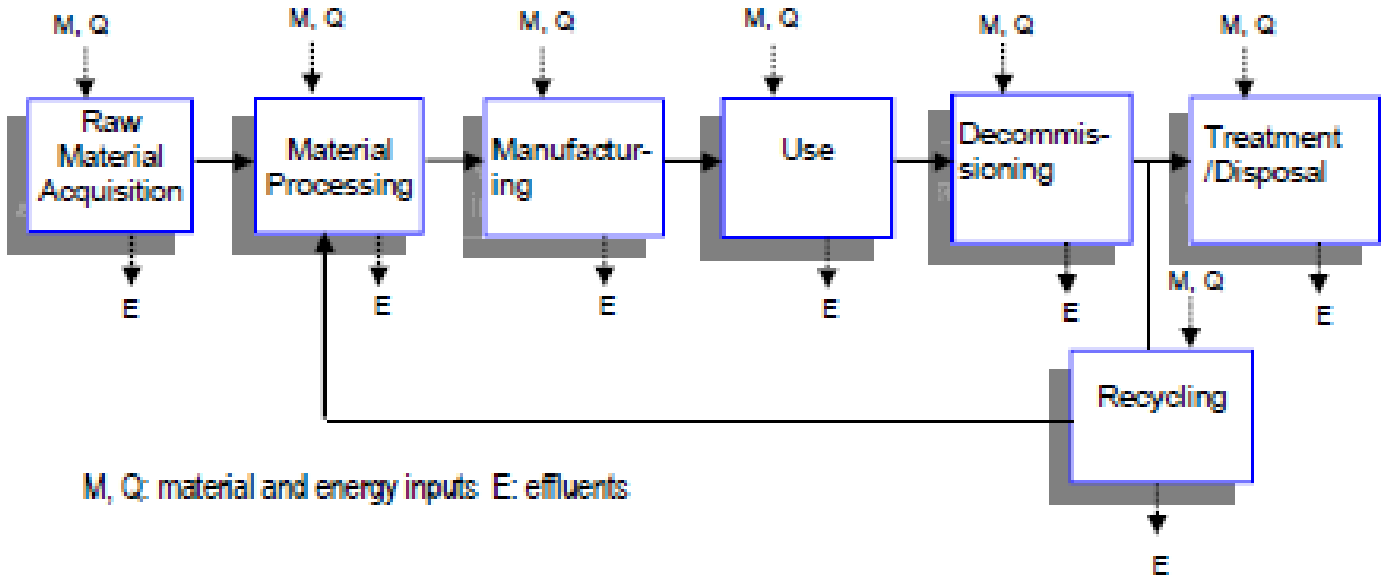
STC: Irradiance 1000W/m2, Module temperature 25ºC, AM=1.5

Mechanical Characteristics	
Solar Cell	Mono-crystalline 125×125mm (5inch)
No. of Cells	72 (6×12)
Dimensions	1580×808×35mm (62.2×31.8×1.4inch)
Weight	15.5kg (34.1lbs.)
Front Glass	3.2 mm (0.13inch) tempered glass
Frame	Anodized aluminium alloy
Junction Box	IP65 rated



http://www.arcmansolar.com/products/44.aspx?cid=7-12-11

LIFE CYCLE ASSESMENT



What is net benefit of a PV product if we consider all the resource and energy we consumed to produce it at all the energy we generated by using it?

LIFE CYCLE ASSESMENT

Indicators:

Radionuclide emissions, nuclear-waste generation, and air-pollutant emissions (NOx, SO2, PM2.5, PM10), greenhouse-gas emissions, cumulative energy demand, acidification potential (AP), ozone depletion potential (ODP), human toxicity, ecotoxicity, (Abiotic) Resource Depletion (for example scarce metals, like silver, indium, tellurium, and gallium, carry considerable uncertainty, Land use and water use……..

To interpret results better, combine indicators as:

-Energy payback time (EPBT),

-Energy return on investment (EROI) or

-Impact mitigation potentials (IMP)

LIFE CYCLE ASSESMENT

Energy Payback Times:

Energy payback time (EPBT) is the length of deployment required for a photovoltaic system to generate an amount of energy equal to the total energy that went into its production. The value of EPBT is dependent on three factors:

- the conversion efficiency of the photovoltaic system;
- the amount of illumination (insolation) that the system receives (about 1700 $kWh/m^2/yr$ average for southern Europe and about 1800 $kWh/m^2/yr$ average for the United States); and
- the manufacturing technology that was used to make the photovoltaic (solar) cells.

http://www1.eere.energy.gov/solar/pv_basics.html

LIFE CYCLE ASSESMENT

Energy Payback Time = $(E_{mat}+E_{manuf}+E_{trans}+E_{inst}+E_{EOL}) / ((E_{agen} / n_G) - E_{O\&M})$

where,

E_{mat} : Primary energy demand to produce materials comprising PV system

E_{manuf} : Primary energy demand to manufacture PV system

E_{trans} : Primary energy demand to transport materials used during the life cycle

E_{inst} : Primary energy demand to install the system

E_{EOL} : Primary energy demand for end-of-life management

E_{agen} : Annual electricity generation

$E_{O\&M}$: Annual primary energy demand for operation and maintenance

n_G : Grid efficiency, the average primary energy to electricity conversion efficiency at the demand side

$$EROI = lifetime / EPBT = T \cdot ((E_{agen}/n_G) - E_{O\&Mr}) / (E_{mat}+E_{manuf}+E_{trans}+E_{inst}+E_{EOL})$$

Life Cycle Inventories and Life Cycle Assessments of Photovoltaic Systems, 2015

LIFE CYCLE ASSESMENT

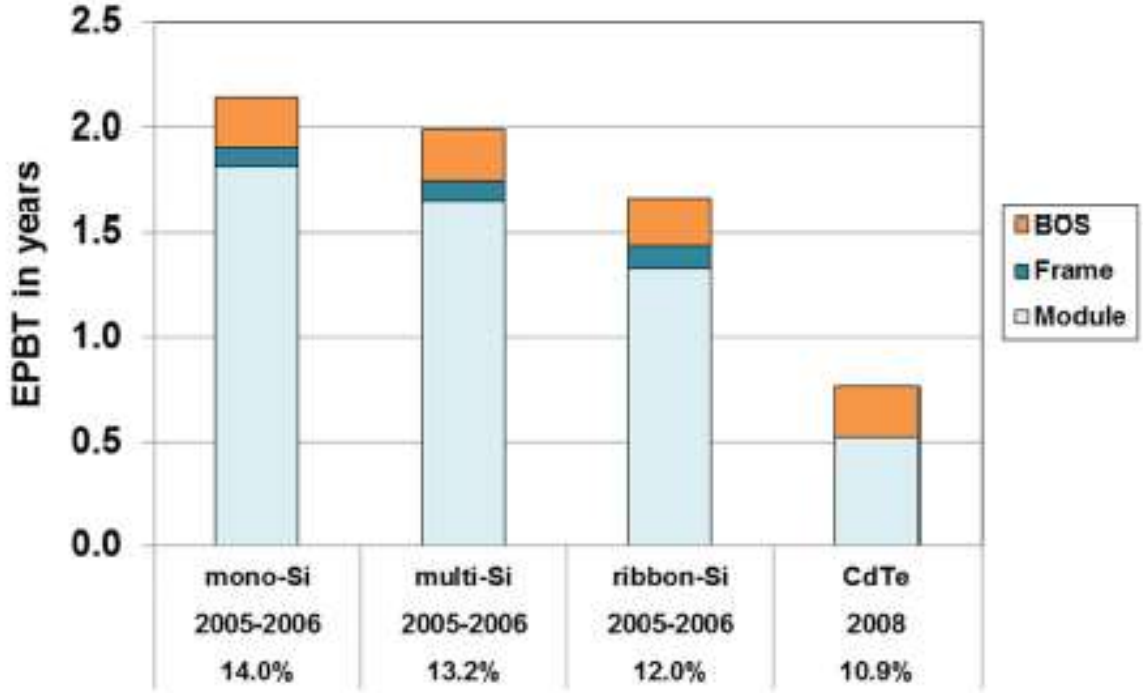


Figure 2: Energy payback time (EPBT) of rooftop mounted PV systems estimated from the currently available LCI data for European production and installation. The estimates are based on Southern European irradiation of 1700 kWh/m2/yr and performance ratio of 0.75. See Tables 5.1.1-5.2.3, 5.4.1, and 5.5.1.1-5.5.1.4 for the corresponding LCI data.

Life Cycle Inventories and Life Cycle Assessments of Photovoltaic Systems, 2015

APPLICATIONS

Central PV Power plants

Lucainena de las Torres PV power plant, Andalusia, Spain (courtesy: MEPSola r)

http://www.pvresources.com, Annual review 2008

APPLICATIONS

REN21 RENEWABLES 2020 GLOBAL STATUS REPORT

Central PV Power plants

APPLICATIONS

Home applications

APPLICATIONS

Stand alone PV system sizing

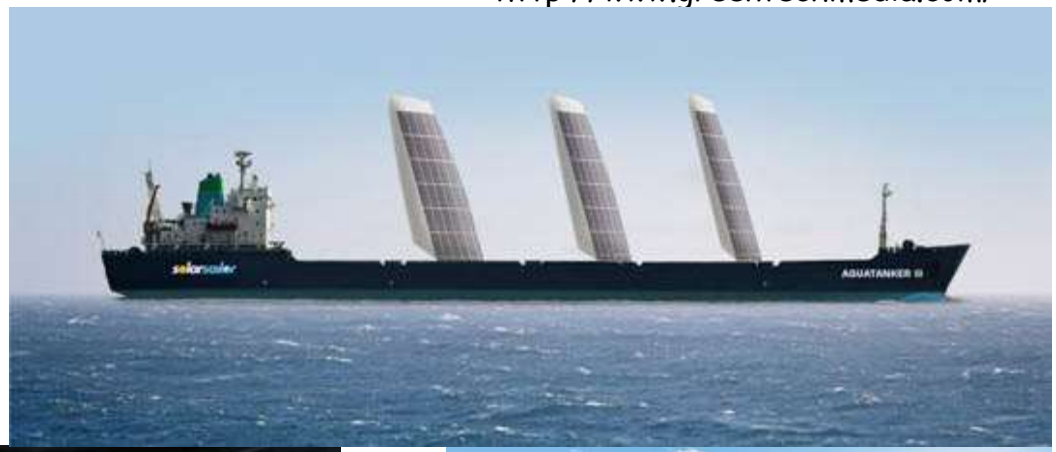
- Determine your power need
  - List power needs of all appliances and lights; calculate weekly (or monthly) AC power consumption by multiplying power (watt) of appliances with working time (hr.) to find energy as (wh orm kwh)
  - Or alternatively read your monthly bill for energy consumption (given as kwh).
- Calculate battery size
  - Decide the number of days for which you want to store energy.
  - Decide battery voltage
  - Divide the energy needs (kwh) to battery voltage to calculate the battery capacity as Amp-hr (also consider losses and inefficiencies etc.)
  - Decide the battery size and calculate the number of batteries needed (also considering how to make the serial and parallel connections)
- Determine panel area
  - Find insolation data
  - Calculate PV area need from the power requirement, insolation data and PV panel characteristics
  - Calculate number of units

*See: dnr.**louisiana**.gov/.../STANDALONE_**PV_Sizing** for details*

APPLICATIONS

Transportation

http://www.greentechmedia.com/



http://www.dasolar.com

http://newspaper.li/solar-car/

APPLICATIONS

Solar forest

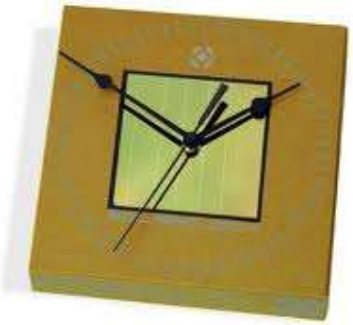
www.treehugger.com

http://inhabitat.com

www.maxiapple.com

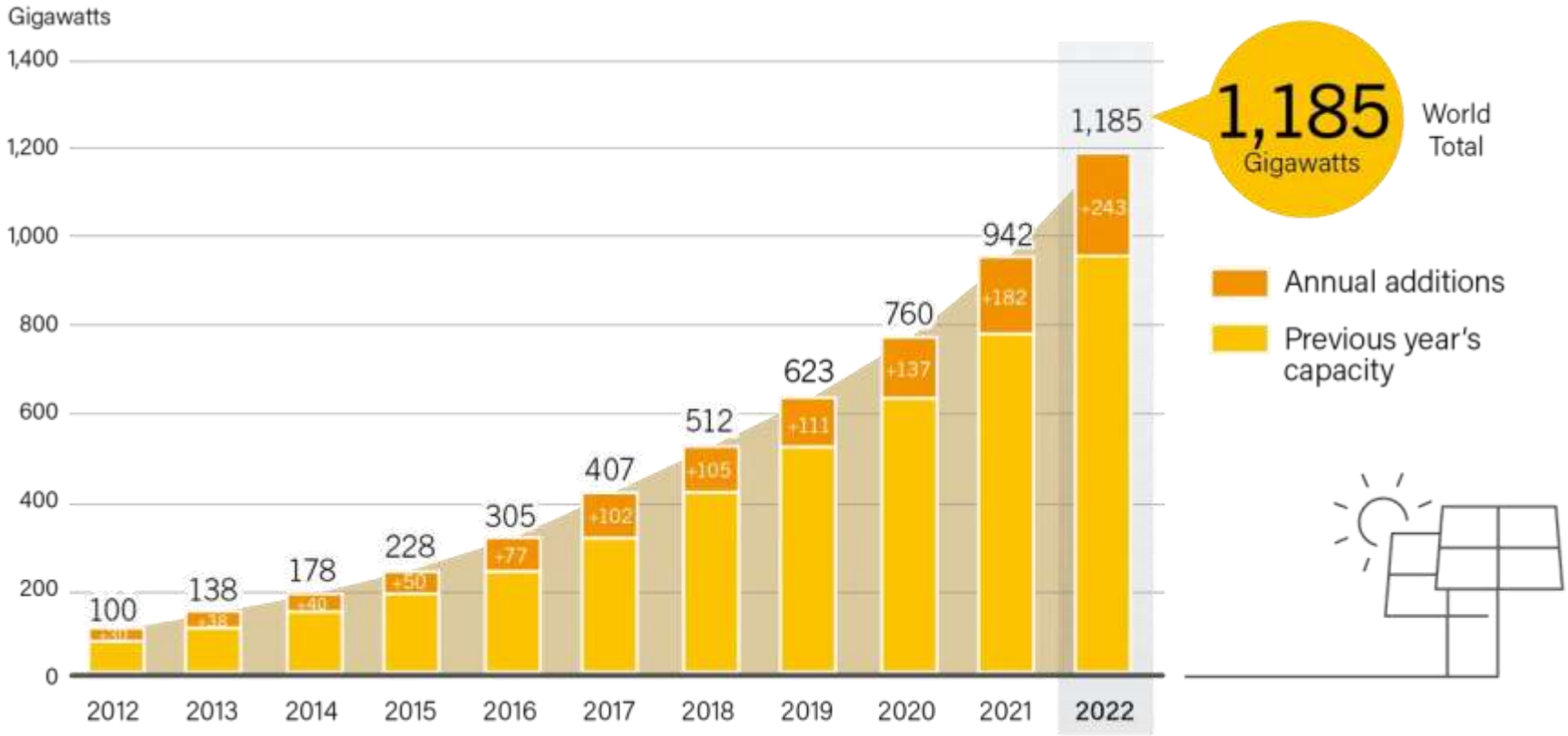
APPLICATIONS

Others



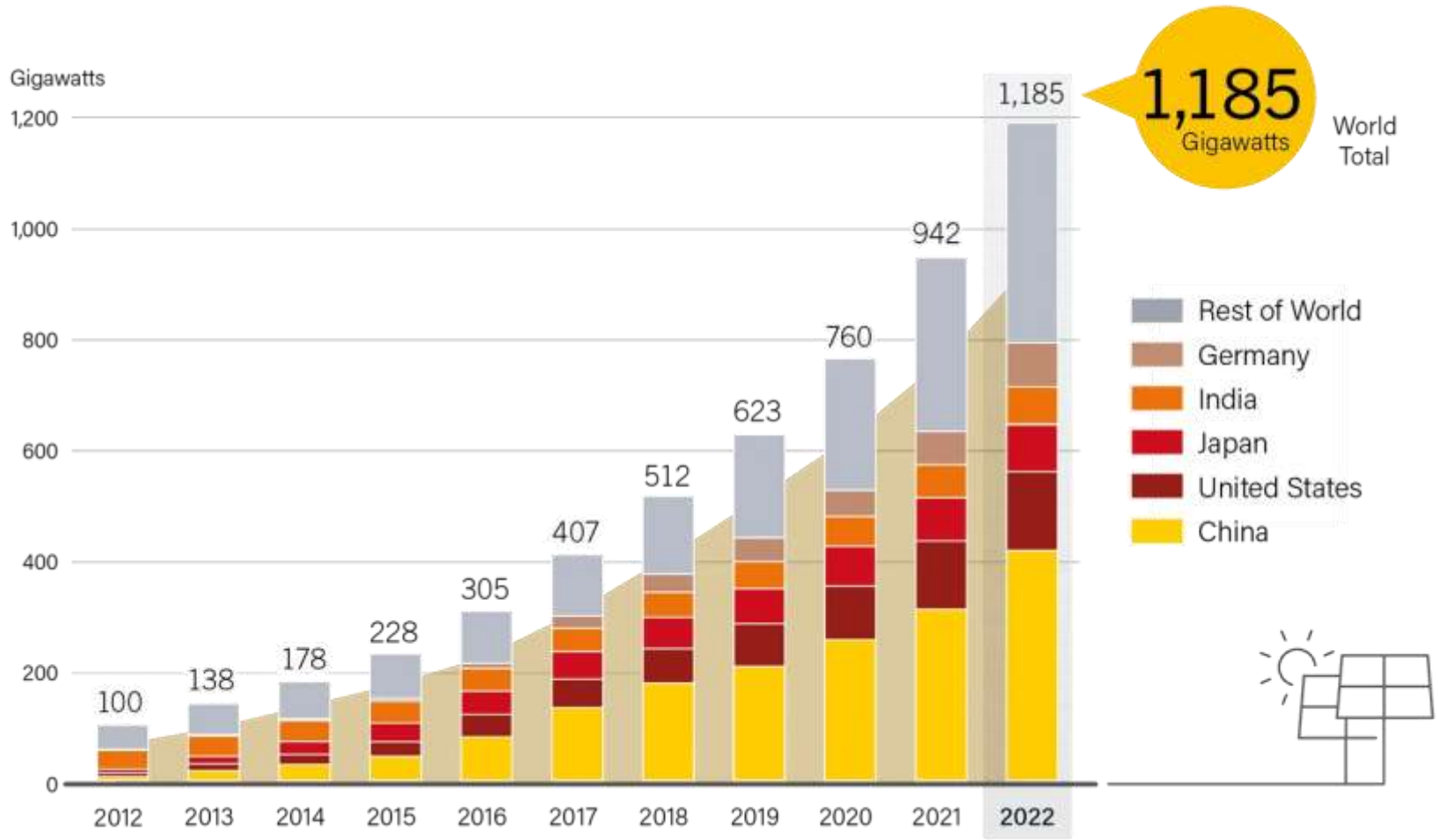
FACTS AND TRENDS

REN21 RENEWABLES 2023 GLOBAL STATUS REPORT



FACTS AND TRENDS

REN21 RENEWABLES 2023 GLOBAL STATUS REPORT



FACTS AND TRENDS

REN21 RENEWABLES 2023 GLOBAL STATUS REPORT

- Solar PV maintained its record-breaking streak, with new capacity increasing 37% in 2022, while global solar production reached an average of 6.2%, up from 5% in 2021.
- For the tenth consecutive year, Asia dominated regionally in new solar PV installations, contributing 64% of the global added capacity in 2022.
- The leading countries for cumulative installed solar PV capacity remained China, the United States, Japan, India and Germany, while the leading markets for per capita capacity remained Australia, the Netherlands and Germany.
- Poland was a new entrant to the top 10 solar PV installers (eighth globally and third in Europe), adding 4.9 GW of capacity, nearly 50% more than its capacity added in 2021.
- Centralised utility-scale solar PV reached a total of 124.8 GW of new installations, driven by tenders and the attractiveness of power purchase agreements. Distributed PV added 115.2 GW and was driven by falling module costs, which made installations more attractive and accessible.
- Although solar PV panel production remains concentrated in China, more countries have strengthened import barriers and incentives for local manufacturing, pioneered by the United States and India.